

INDIAN INSTITUTE OF REMOTE SENSING
Indian Space Research Organisation
Department of Space, Government of India



Seeing the Unseen

The Mysteries of Synthetic Aperture Radar from Space

By

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About RADAR Development

RADAR

RAdio **D**etection **A**nd **R**anging

History

Neither a single nation nor a single person holds the responsibility for inventing radar technology. The knowledge about “Radar” is an accumulation of many developments and improvements of many scientists from several nations took part in parallel.

1904: The German engineer **Christian Hulsmeyer** invents the telemobiloscope for a traffic monitoring on the water in poor visibility. First practical radar test.

1922: The American electrical engineers **Albert H. Taylor** and **Leo C. Young** of the Naval Research Laboratory (USA) locate a wooden ship for the first time

1930: Lawrence A. Hyland of the Naval Research Laboratory, locate an aircraft for the first time

1931: A ship equipped with radar. As antennae are used parabolic dishes with horn radiators

1940: Different radar equipments were developed in the USA, Russia, Germany, France, and Japan

Military and surveillance radars undergone a major development during second world war.

Sir Robert Watson Watt (1892-1973) considered by many **‘the inventor of radar’**



Remote Sensing

Based on EM Spectra used

- Optical
- Infrared
- Thermal
- Microwave

Based on Source of Energy

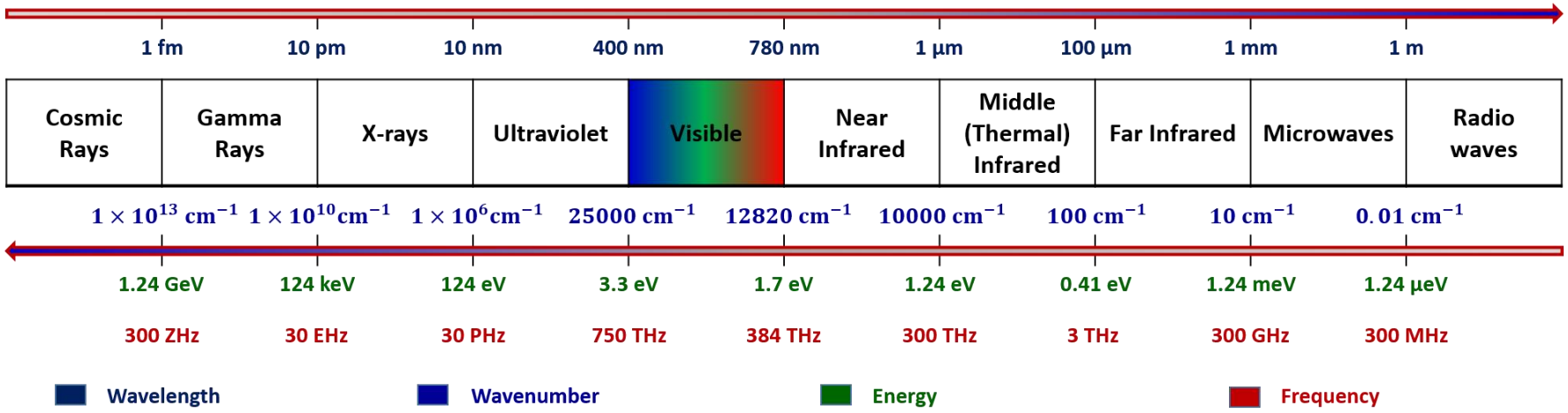
- Active
- Passive

Based on Number of Bands

- Panchromatic
- Multi-spectral
- Hyperspectral
- Ultra spectral

Based on the Platform

- **Space borne:** Satellite based
- **Airborne:** Aircraft based, Balloon based, etc.
- **Ground based (In-situ):** AWS, Instruments etc.



Band Name	Frequency / Wavelength	Band Name	Frequency / Wavelength
Ka	27.0 - 40 GHz / 1.11 - 0.75 cm	C	4.0 – 8.0 GHz / 7.5 - 3.75 cm
K	18 - 27.0 GHz / 1.67 - 1.11 cm	S	2.0 – 4.0 GHz / 15.0 - 7.5 cm
Ku	12.0 - 18 GHz / 2.5 - 1.67 cm	L	1.0 – 2.0 GHz / 30 - 15 cm
X	8.0 - 12.0 GHz / 3.75 - 2.5 cm	P	0.3 – 1.0 GHz / 100 - 30 cm

Why we use microwave in remote sensing?

- Microwave signals are **complex** in nature
- Can give very **high precision** measurements (unto a scale of mm order)
- Can be conducted in **Active** and **Passive** mode
- Not dependent on **Sun illumination** (Active)
- Not dependent on **Weather conditions** (rainy, foggy, cloudy, Smokey, dusty etc.)
- Minimally affected by **Atmosphere**
- Passive microwave remote sensing generally applied from **space** not from **airborne** platform
- Active microwave remote sensing can be performed from **airborne** and **space borne** platforms

Interaction of Microwaves with Different Media

Rain Cloud Soil Vegetation Atmosphere

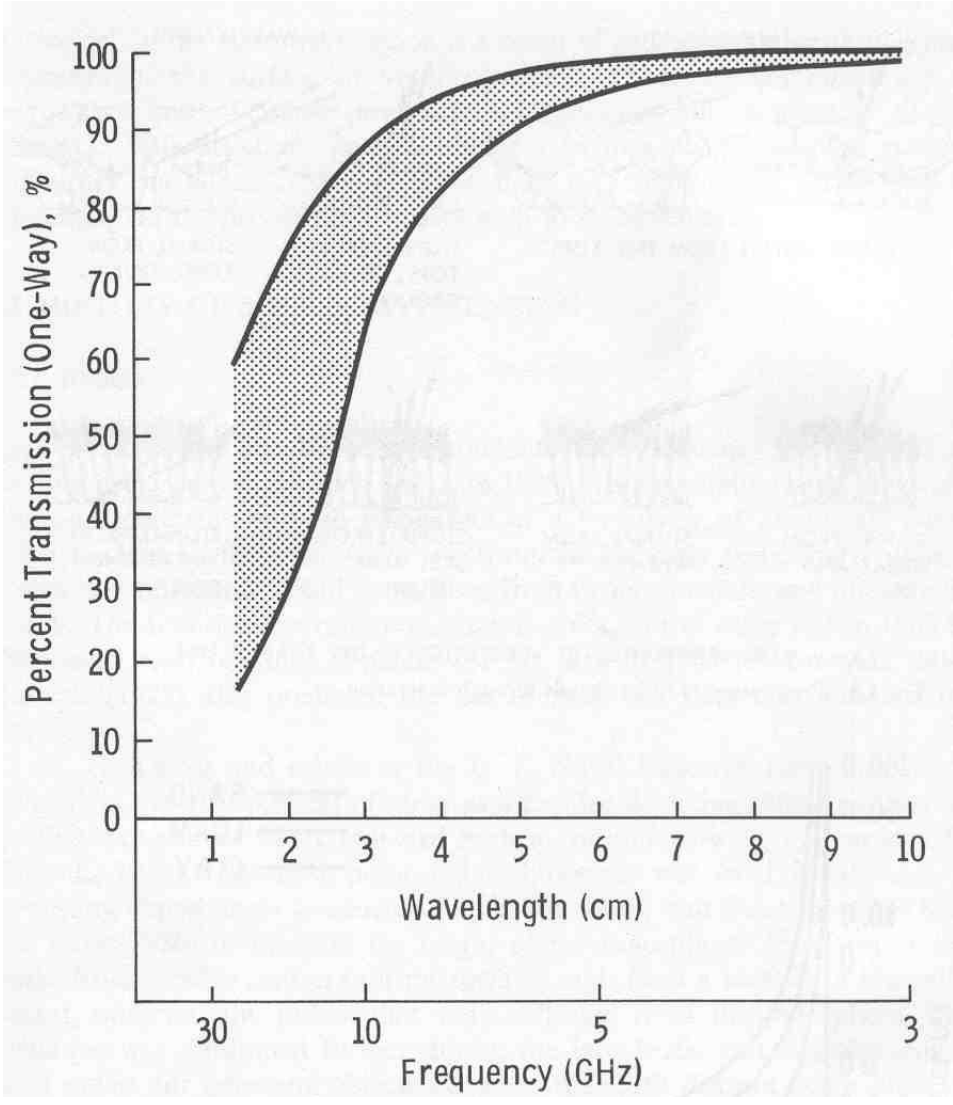
Effect of Rain on Microwaves with change in wavelength/frequency

Rain has greater effect on microwave than clouds

- Effect is negligible for λ above 4 cm
- λ below 2 cm are affected, when rain is mild to string
- λ below 1 cm are affected, when rain is extremely intense

Response of Water on Microwaves

- Smooth surface of standing water reflects almost all incident radiation away from the sensor- **lower backscatter**
- Turbulent surface of water produces **more backscatter** because of **rough surface**
- Standing water under vegetation produces **high backscatter** (corner reflection backscatter between surface water and stem)



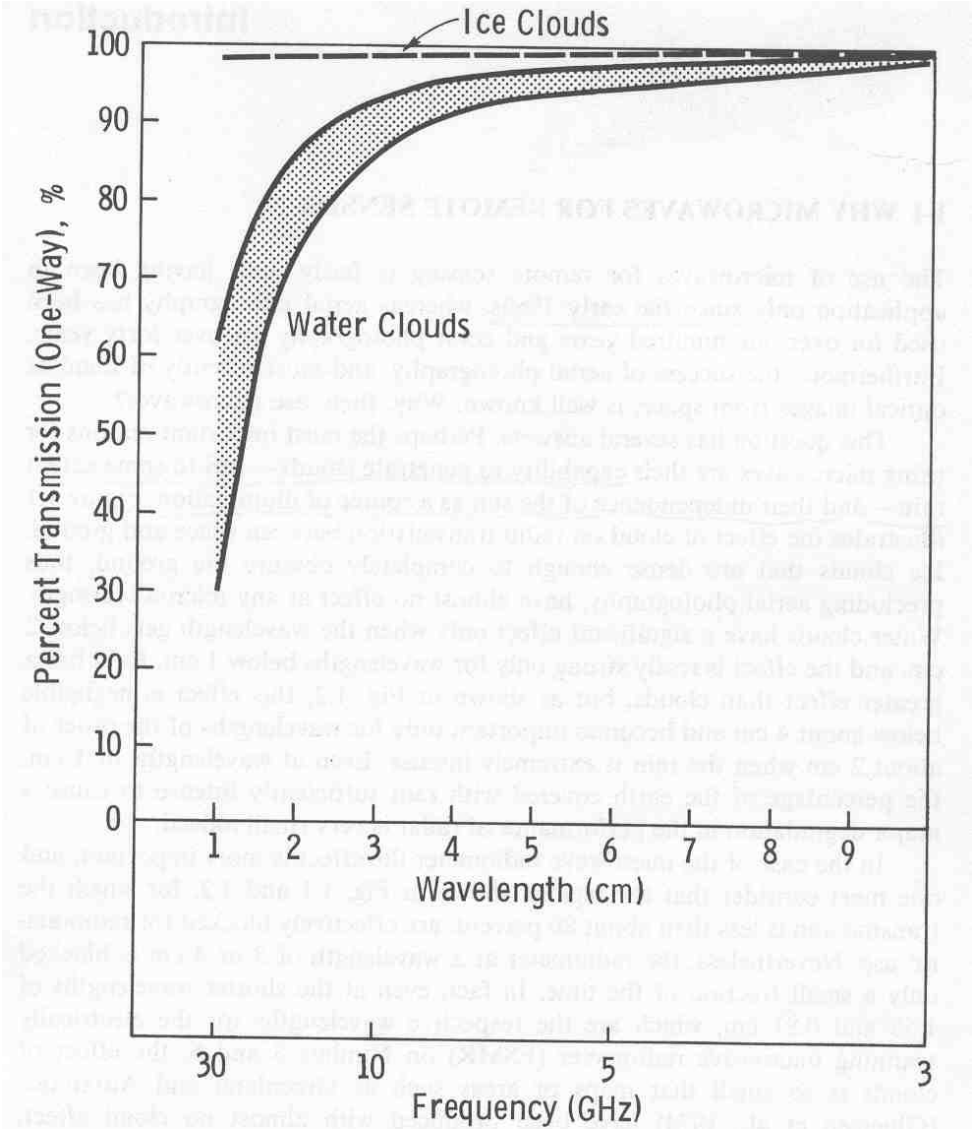
Interaction of Microwaves with Different Media

Rain Cloud Soil Vegetation Atmosphere

Effect of Cloud on Microwaves with change in wavelength/frequency

Microwave can penetrate cloud

- **Ice clouds** are dense enough to completely obscure the ground; have almost **no effect** at any λ in microwave region
- **Water clouds** have **significant effect** only when the λ gets below 2 cm; and the effect is **really strong** for λ below 1 cm.



Interaction of Microwaves with Different Media

Rain | Cloud | Soil | Vegetation | Atmosphere

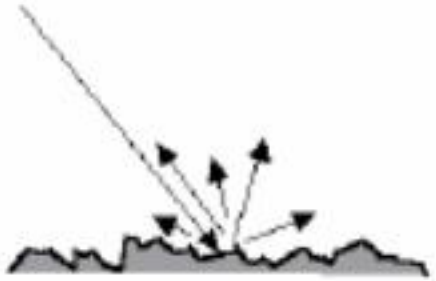
Interaction of Microwaves with Soil

Microwave can penetrate soil

- **Dry Soil:** Some of the incident radar energy is able to penetrate into the soil surface, resulting in less backscattered intensity
- **Wet Soil:** The large difference in electrical properties between **water and air** results in **higher backscattered** radar energy
- **Flooded Soil:** Radar is **specularly reflected** off the water surface, resulting in **low backscattered** intensity. The flooded area appears **dark** in the SAR images



Dry soil



Wet soil



Flooded soil

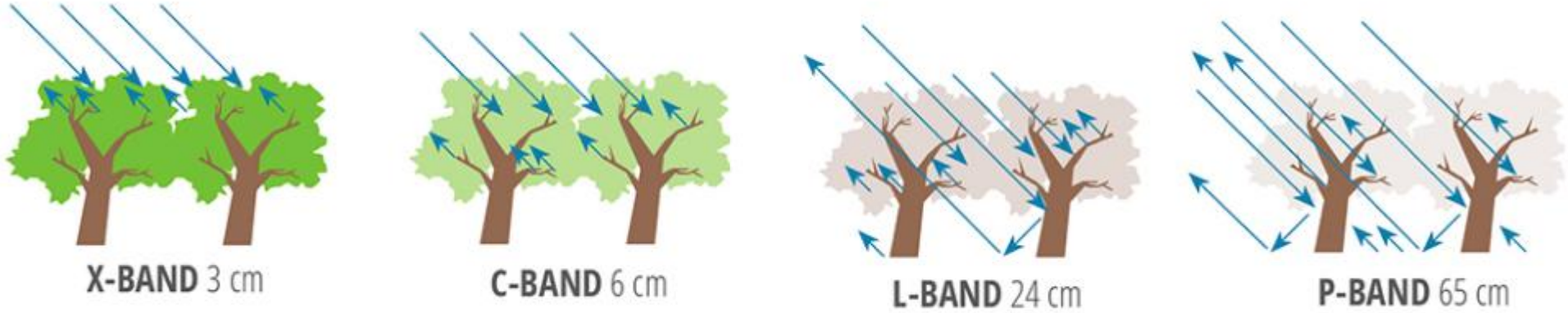
Interaction of Microwaves with Different Media

Rain | Cloud | Soil | Vegetation | Atmosphere

Interaction of Microwave with Vegetation

- **Volume scattering** is related to multiple scattering processes within a medium, such as vegetation canopy (forest or crop)
- **Volume Scattering:** The energy is **depolarized** when scattered multiple times (needle to stem to trunk to needle) within a diffused volume.
- The **intensity** of volume scattering depends on the physical properties of the volume (variations in dielectric constant) and the characteristics of the radar (wavelength, polarization and incidence angle)

- Influenced by the following biophysical parameters-
  - Canopy water content
  - Vegetation type
  - Biomass
  - Canopy structure
- The amount of energy received also depends upon–
  - Whether or not canopy components **depolarize** the signal, how far the signal penetrates, and whether it eventually interacts with the ground soil surface.



Longer the wavelength, deeper the penetration, more volume scattering

Source: https://www.earthdata.nasa.gov/learn/backgrounders/what-is-sar

Interaction of Microwaves with Different Media

Rain Cloud Soil Vegetation Atmosphere

Effects of Atmospheric propagation on Microwave

Troposphere (<50 km)

- Refractive index, $\mathrm{n}(h) > 1$ in troposphere is positive and decreases with height. Resulted into reduction in velocity, similar to an elongation of wave path.
- The following integral gives the path elongation for the troposphere crossing according to an average look angle (θ) and according to co-index $N(h)$ (Refractivity)

$$\Delta l_{tropo} = \frac{10^{-6}}{cos\theta} \int_0^{h_{max}} N(h)\, dh$$

$$N(T, P, e) = 77.6\frac{P}{T} - 5.6\frac{e}{T} + 0.375 \times 10^6 \frac{e}{T^2}$$

Where T = temperature in Kelvin
P = atmospheric pressure
e = partial water pressure

Ionosphere (50-1000 km)

Ionosphere Layer	Height	Diurnal ρ	Nocturnal ρ
Layer D	50-90 km	$< 10^9$	~ 0
Layer E	90 – 150 km	$\sim 10^{11}$	$\sim 10^9$
Layer F	> 150 km	$\geq 10^{12}$	$\geq 10^{11}$

Where ρ is the electron density per meter cube.

Two-way ionospheric delay: $\Delta S = 2 \times K \times \frac{TEC}{f^2\, cos\alpha_{offnadir}}$

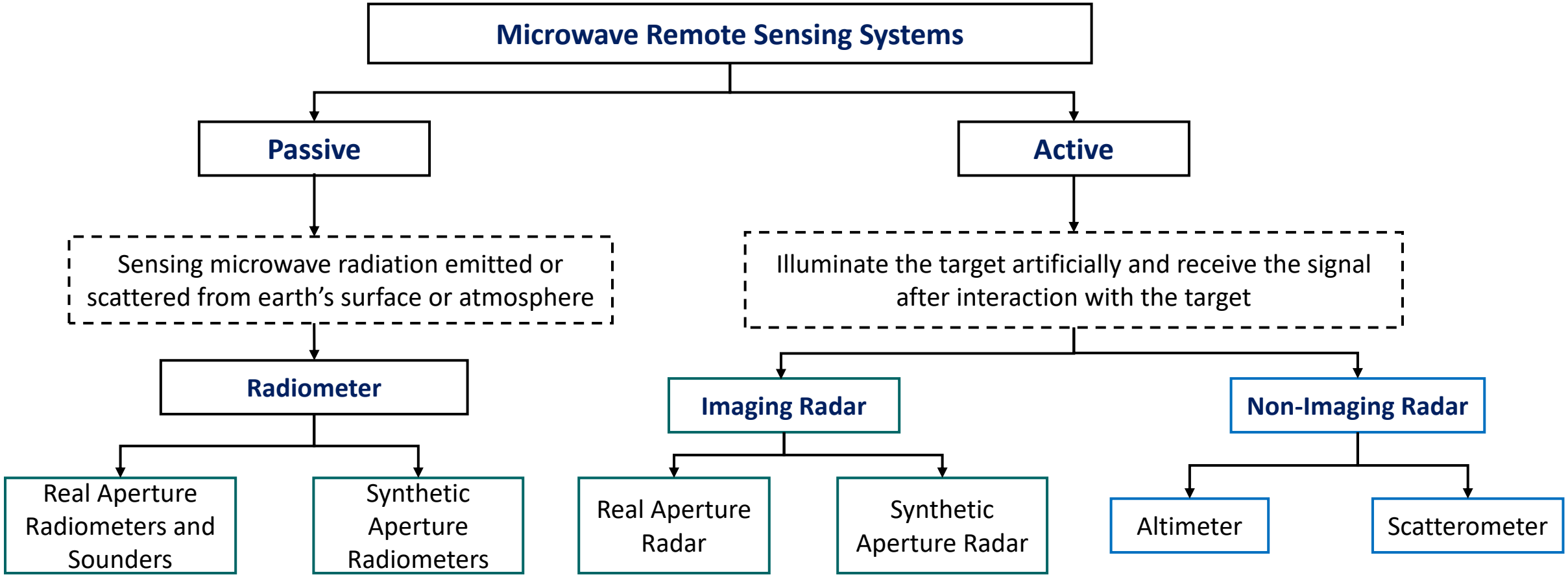
$\mathrm{K} = \text{refractivity constant} = -40.28 \frac{\mathrm{m}^3}{\mathrm{s}^2}$

$\mathrm{f} = \text{frequency of the L} - \text{band} = 1.276\ \mathrm{GHz}$

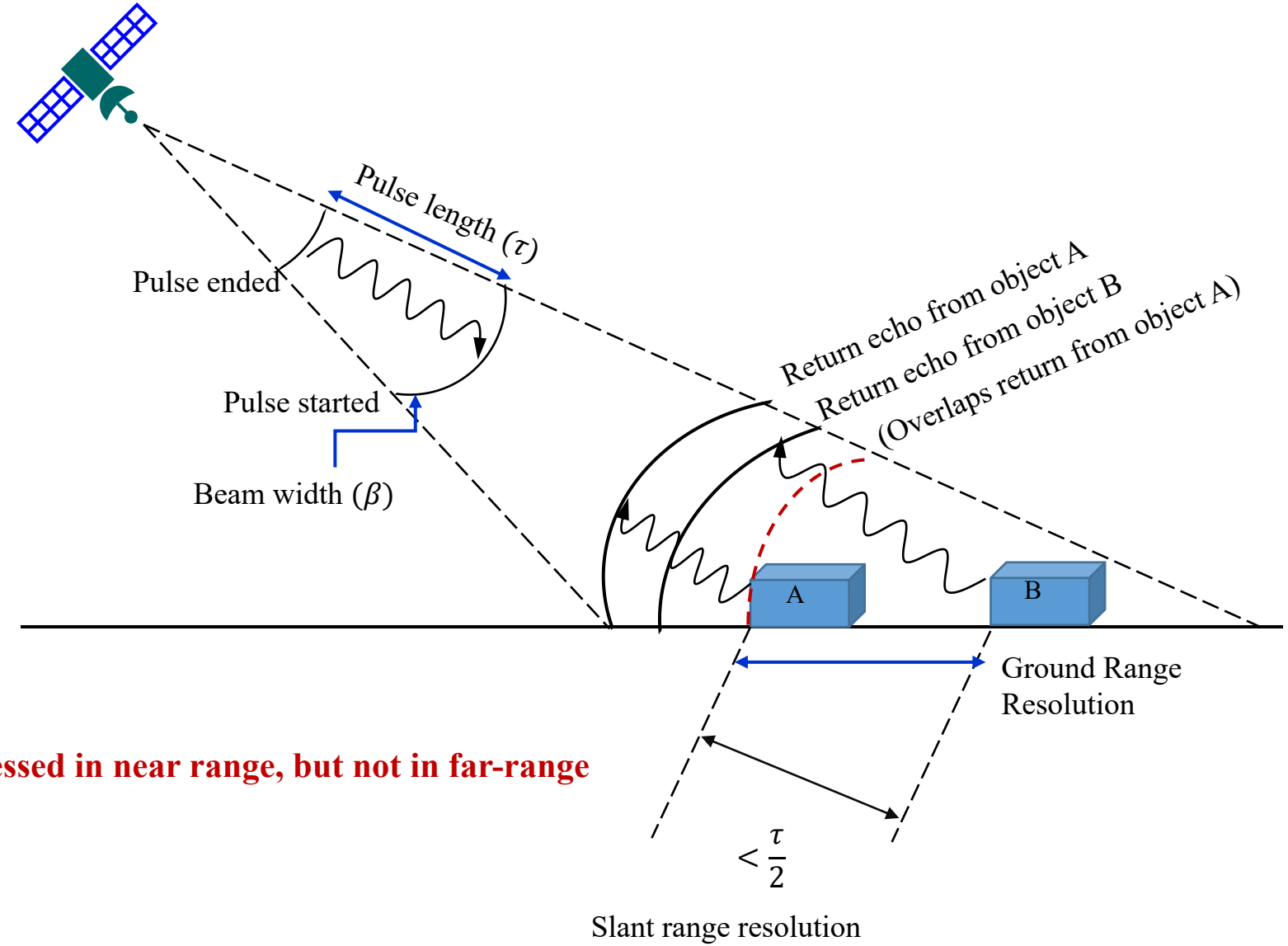
$\alpha_{\mathrm{offnadir}} = \text{Satellite Offnadir angle} = 34.3\ \text{degrees}$

$\mathrm{TEC} = \text{total electron content}$

Microwave Remote Sensing Systems

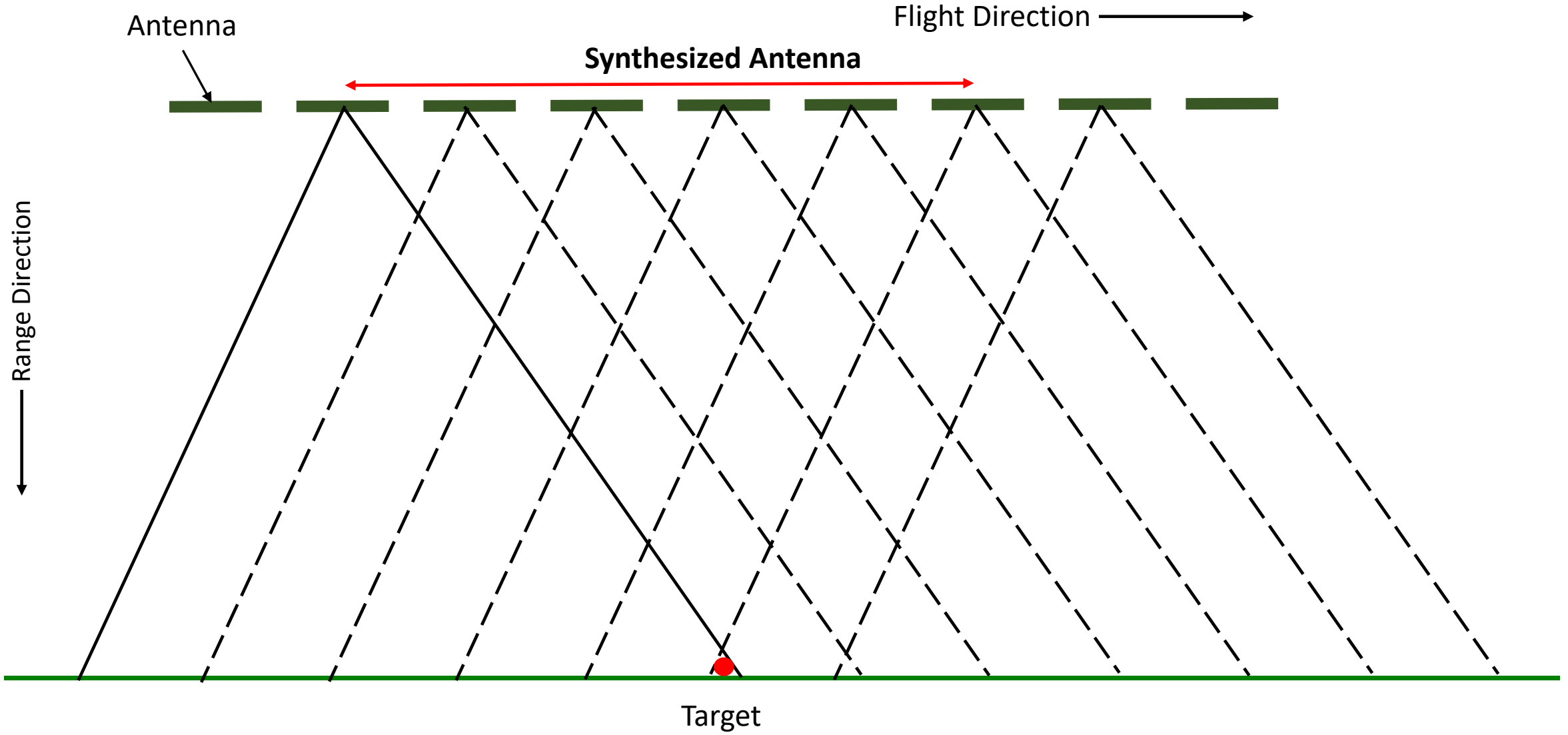


Radar in Remote Sensing (Side Looking)



Compressed in near range, but not in far-range

Formation of Synthetic Aperture



Radar Image Parameters

Radar return and Brightness of Radar Image depends on many parameters –

- **System Parameters/Properties (SP)**
  - Frequency /wavelength
  - Polarization
  - Speckle
  - Viewing Geometry
  - Spatial Resolution

  Frequency /wavelength, Polarization, Speckle → **Radiometric aspects**

  Viewing Geometry, Spatial Resolution → **Geometric aspects**

- **Terrain Parameters/Properties (TP)**
  - Surface geometry
  - Dielectric properties
  - Surface roughness
  - Volume Scattering

  Surface geometry → **Geometric aspect**

  Dielectric properties → **Radiometric aspect**

  Surface roughness, Volume Scattering → **Structural aspects**

SP: Wavelength

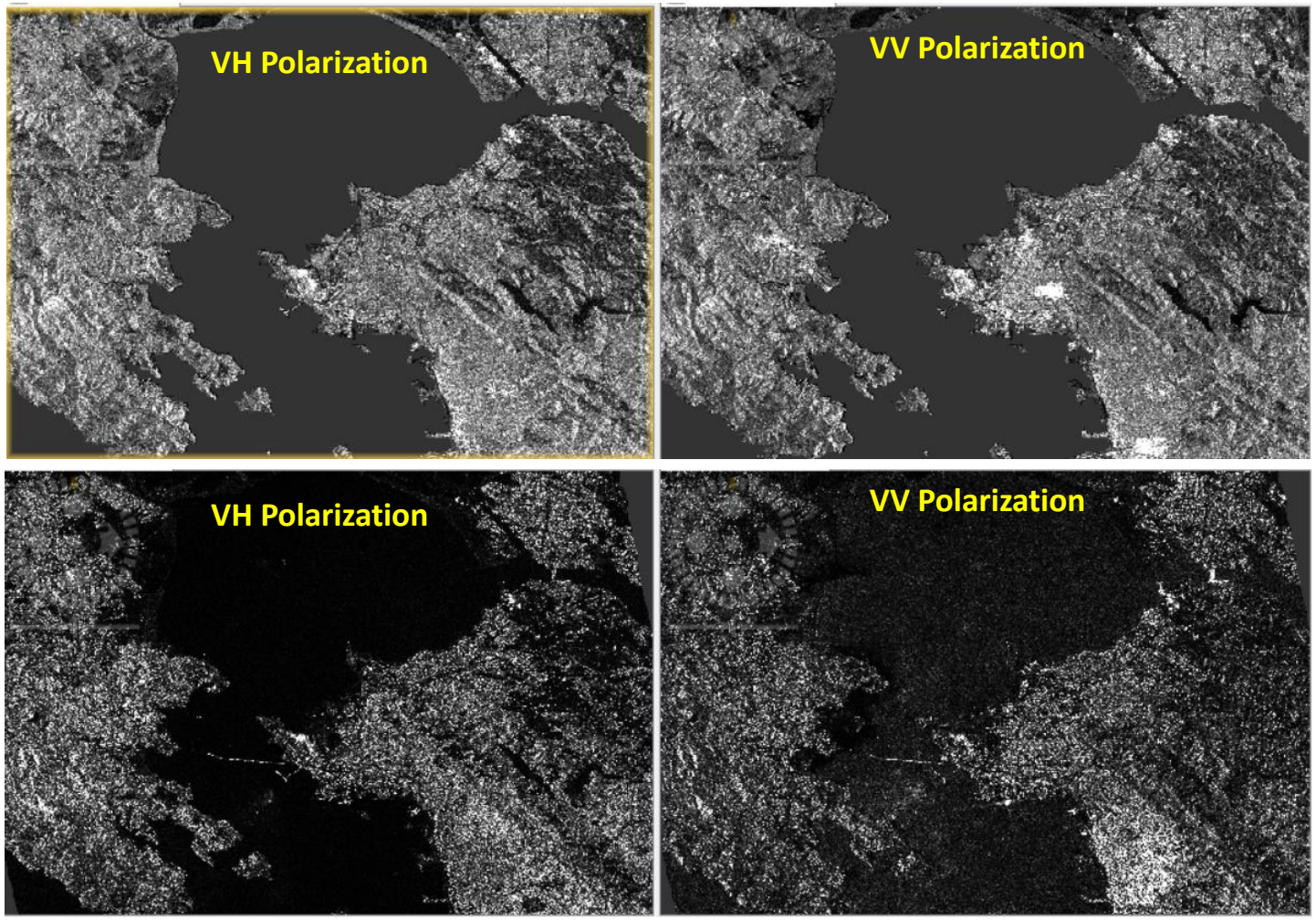
Important Points:

- Wavelength is a measure of surface scale / roughness
- Longer wavelengths penetrates more than shorter wavelengths
- Rough surface scatters more energy towards sensors
- Surface roughness depends on wavelength

System factors:

- **High wavelengths (Low frequencies)**
  - More difficult processing
  - Need larger antennas and feeds
  - Simpler electronics
- **Low wavelengths (High frequencies)**
  - Need more power
  - More difficult electronics
  - Good component availability at X-band

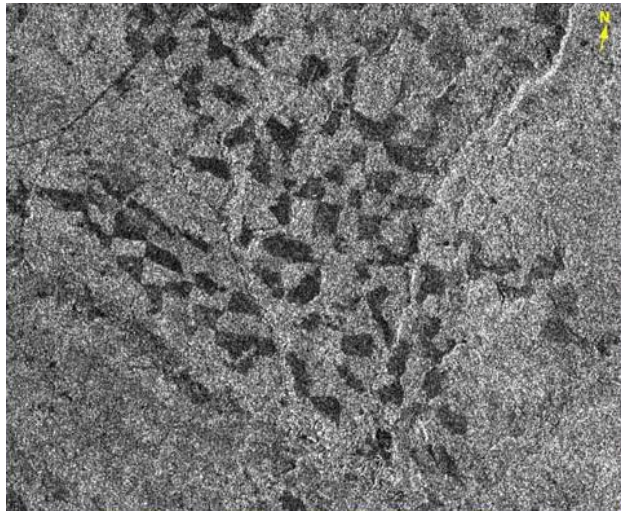
Sentinel-1 (C-band)



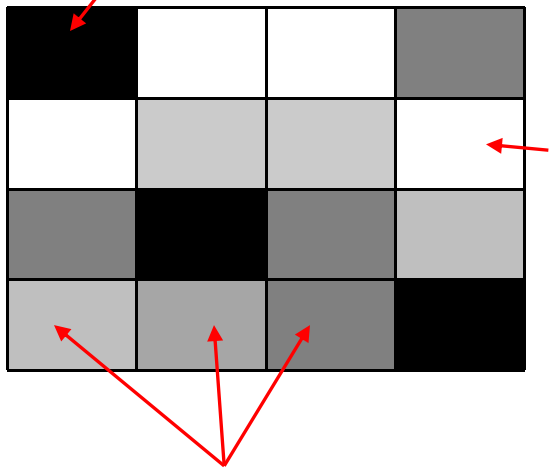
ALOS-2 (L-band)

SP: Speckle

- In full resolution SAR imagery, speckle appears as grainy ‘slat and pepper’ texture
- In a ground resolution cell the individual scatterers differ in their delay time but not recorded as different features because of the resolution limitation, gives resultant signal, the summation of signals from individual scatterrers which differ in phase. This phase difference causes the interference and due to this the resultant may vary from zero to the absolute sum of the individual signals.
- Speckle is the random noise (inherently exists) in Radar images due to interference phenomenon.
- Caused by random constructive and destructive interference from the multiple scattering returns
  - Signals in-phase cause constructive interference and appears bright
  - Signals out of phase cause destructive interference and appears dark
- Speckle noise occupies a wider dynamic range than the scene content itself.



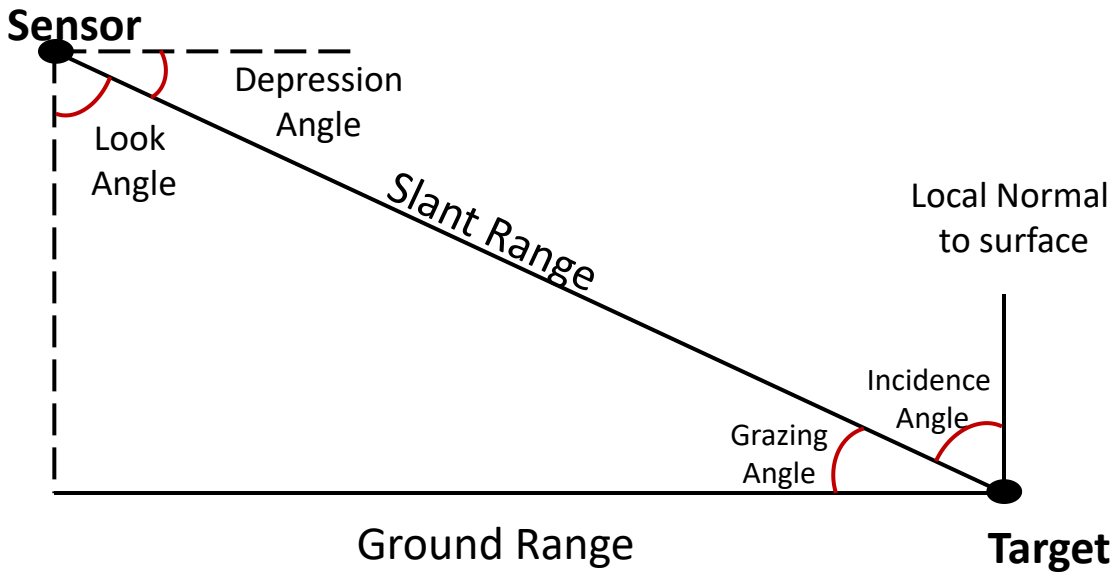
Destructive Interference

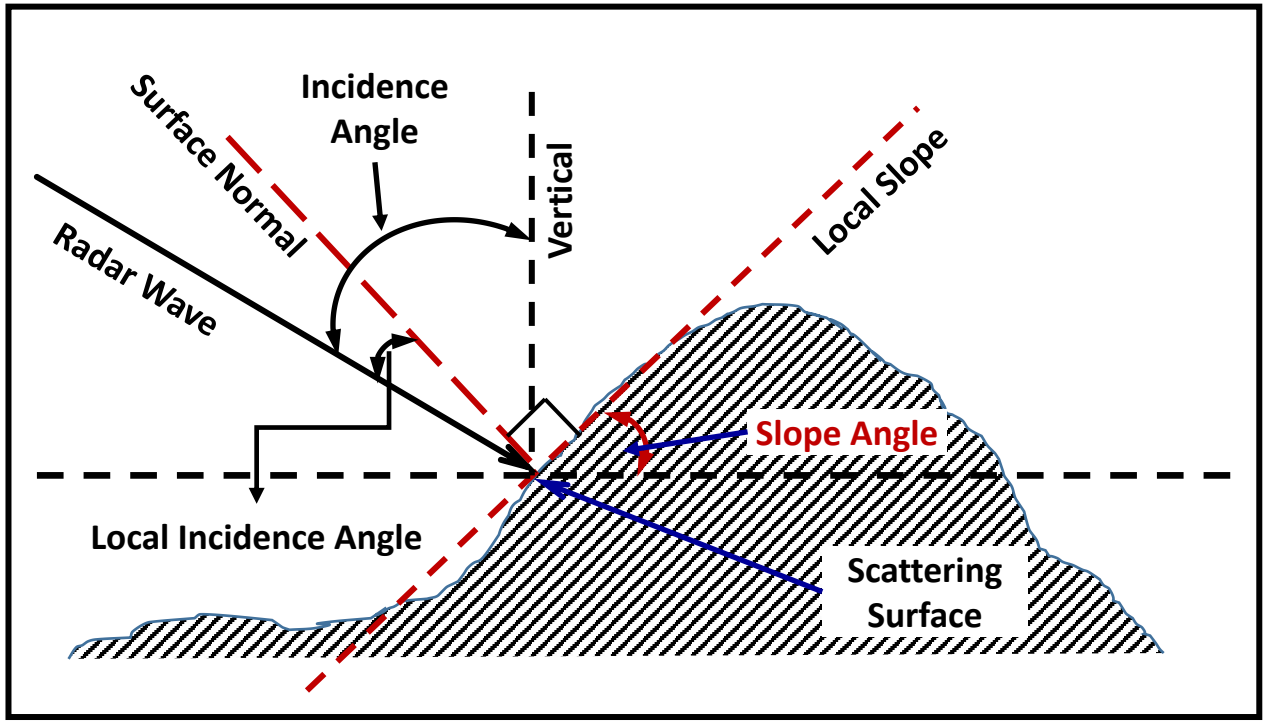


Constructive Interference

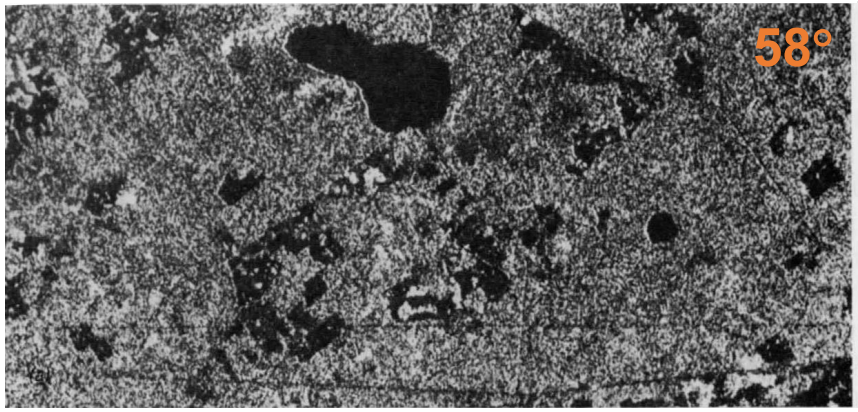
Varying degree of Interference
(between constructive and destructive)

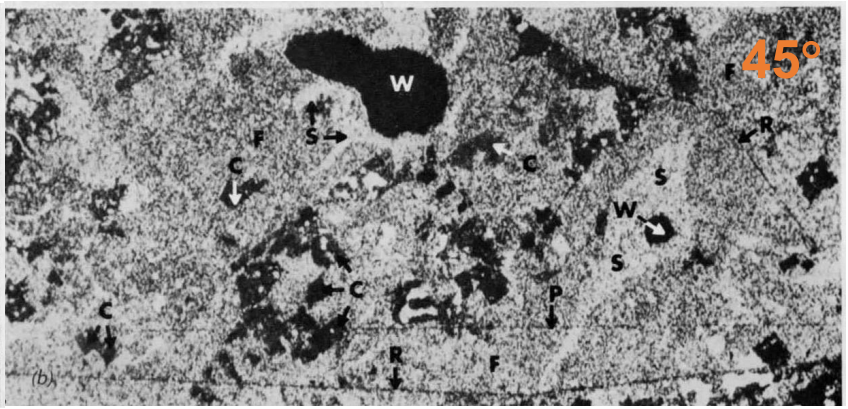
SP: Viewing Geometry/Angles

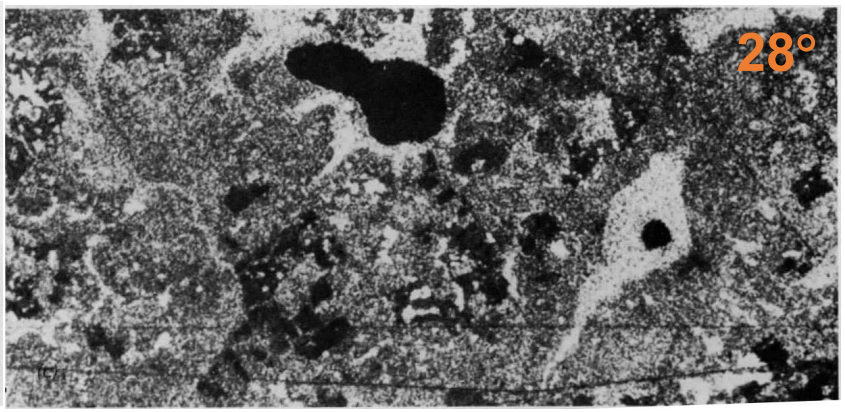




Acquisitions at Different Incidence Angles

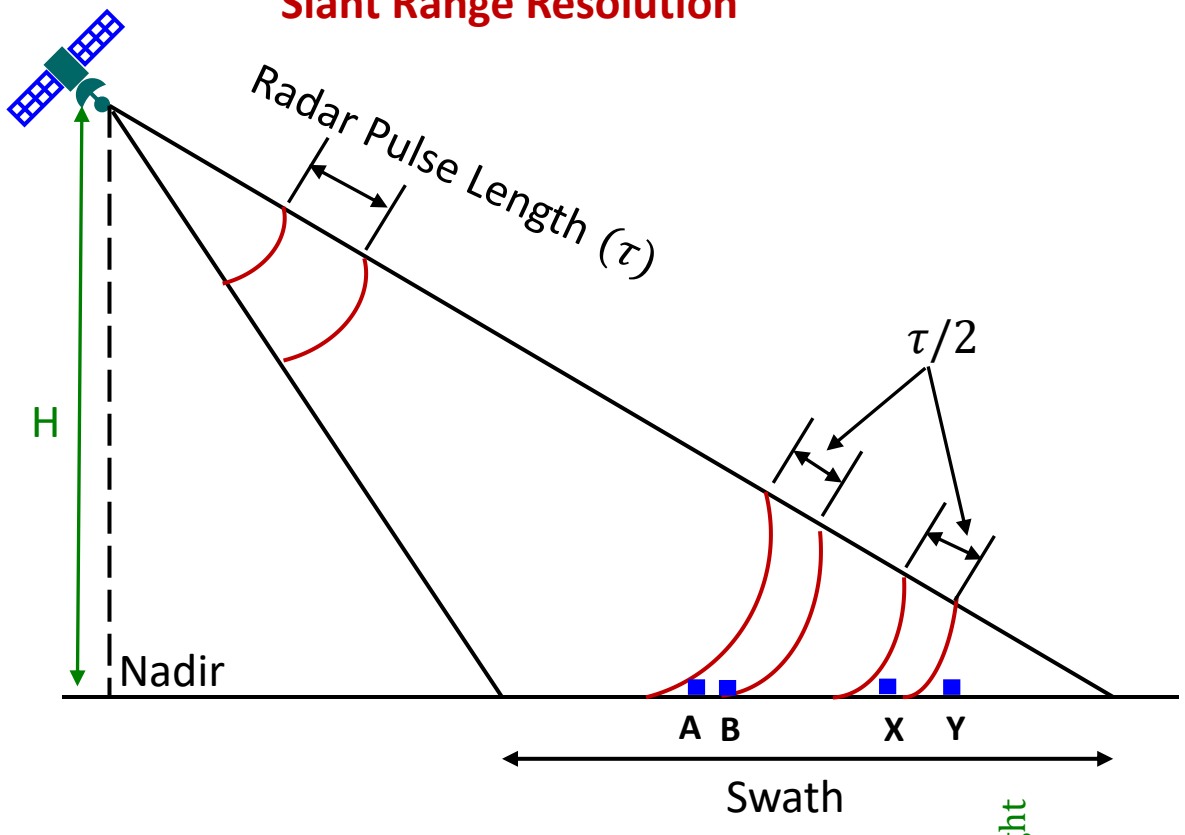






SP: Spatial Resolution (Range)

Slant Range Resolution

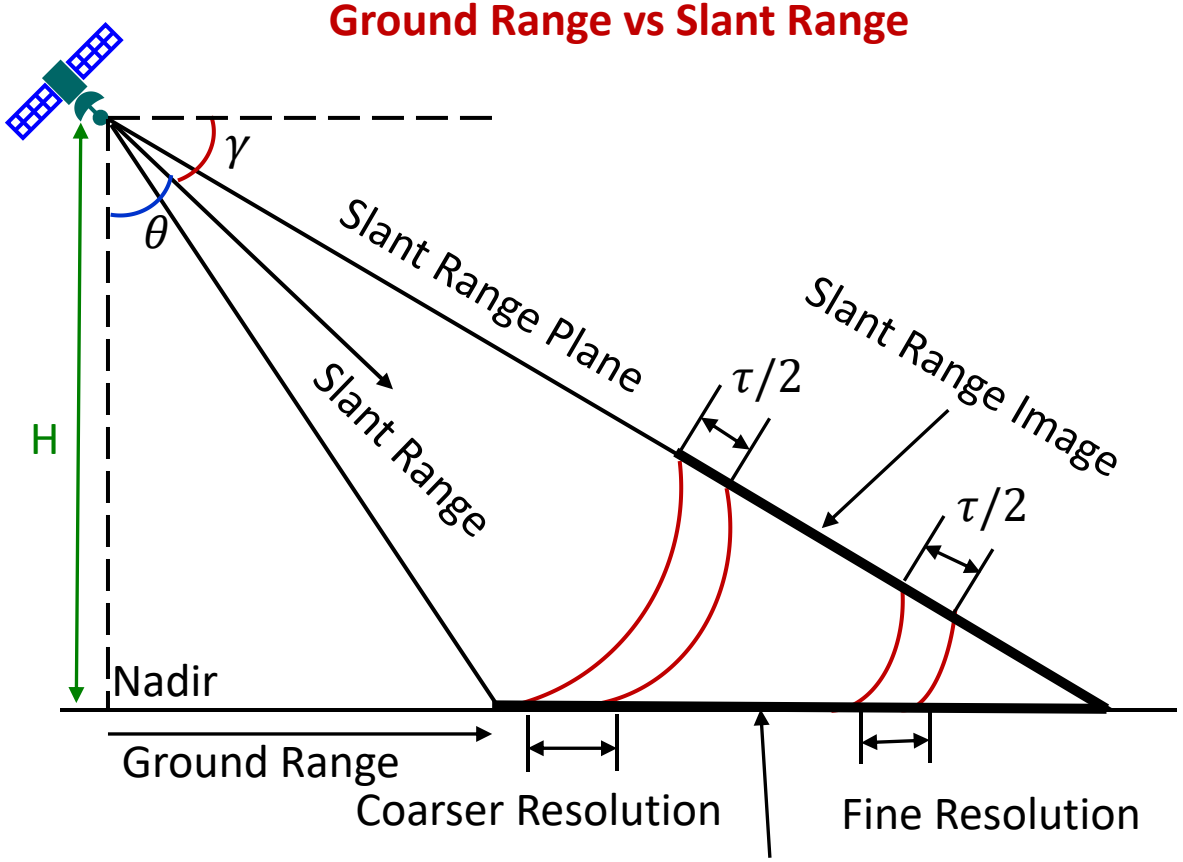


$$Range\ Resolution\ (R_r) = \frac{c\tau}{2}$$

$$Ground\ Range\ Resolution\ (GR_r) = \frac{c\tau}{2\sin\theta}$$

Independent of Height

Ground Range vs Slant Range



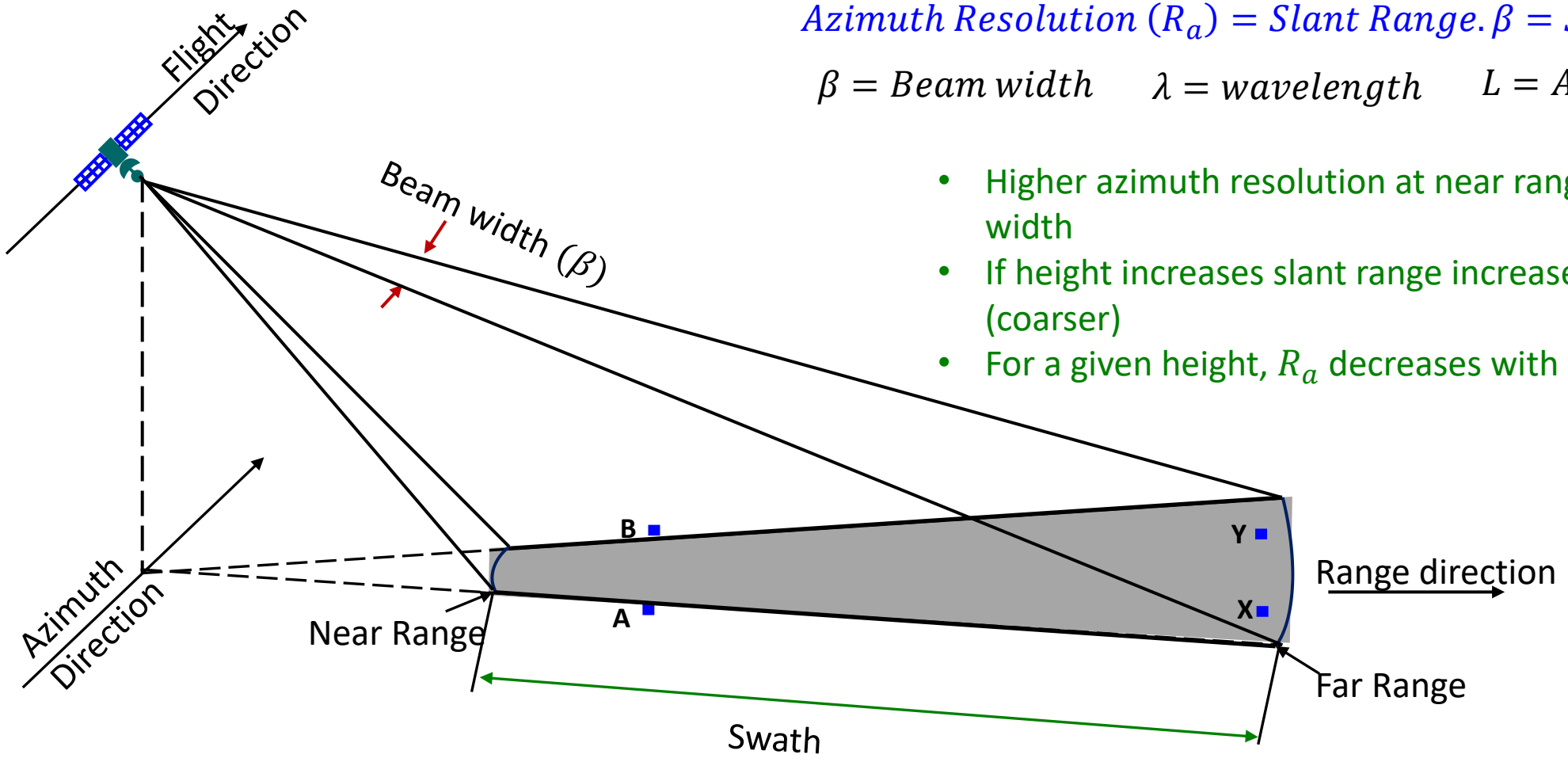
Ground Range Image

γ	80°	60°	40°	20°
θ	10°	30°	50°	70°
Slant range resolution (m)	10	10	10	10
Ground range resolution (m)	29	16	12	10.1

SP: Spatial Resolution (Azimuth)

$$Azimuth\ Resolution\ (R_a) = Slant\ Range.\beta = Slant\ Range.\frac{\lambda}{L}$$

$\beta = Beam\ width \quad \lambda = wavelength \quad L = Antenna\ Length$



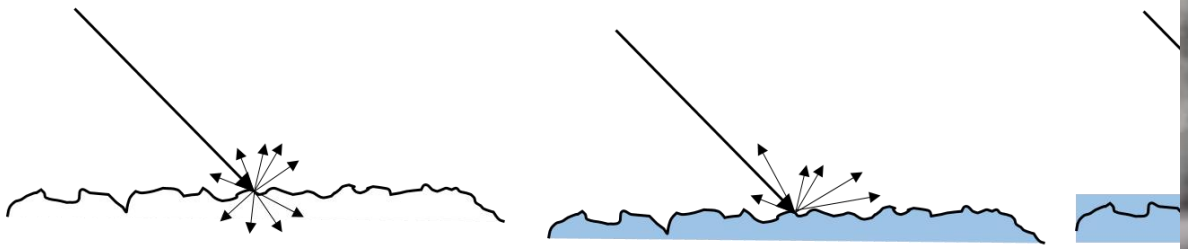
- Higher azimuth resolution at near range, and for narrower beam width
- If height increases slant range increases and R_a decrease (coarser)
- For a given height, R_a decreases with increase in ground range

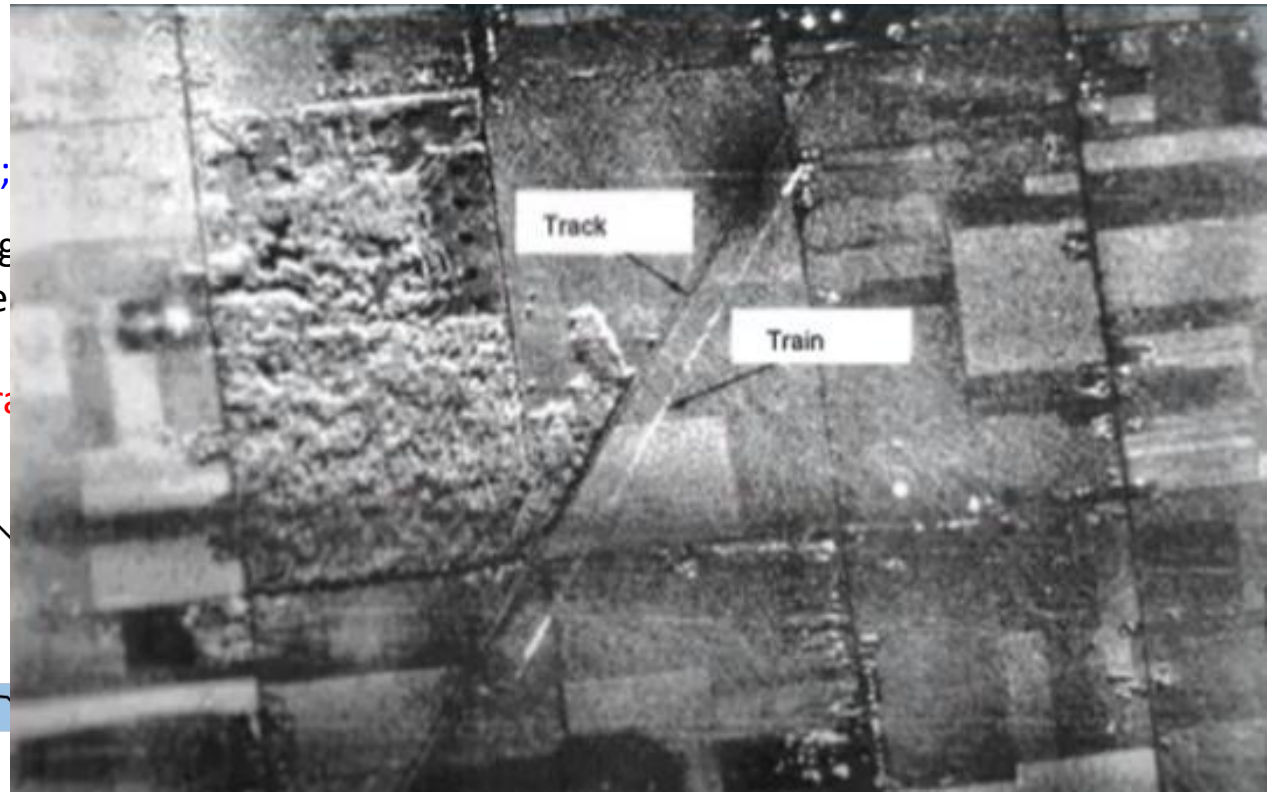
In order to get 25 m resolution with L-band (25 cm wavelength) and 100 km distance from a target, we need 1 Km long antenna

TP: Dielectric Properties

- A measure of a material's electrical characteristics is the complex dielectric constant, defined as a measure of the ability of the material to conduct electricity.
- Dielectric constant: ability of the material to get depolarized when EM field is applied. This material depolarization in turn is related with the amount of polarization of the interacting energy i.e. if the material has very high dielectric constant, it will depolarize the incident EMR better and hence the intensity of received signal will also be more which gives rise to brighter tone.
- Most significant parameter influencing a material's dielectric constant is its moisture (water) content.
- Dielectric constant for some common targets is

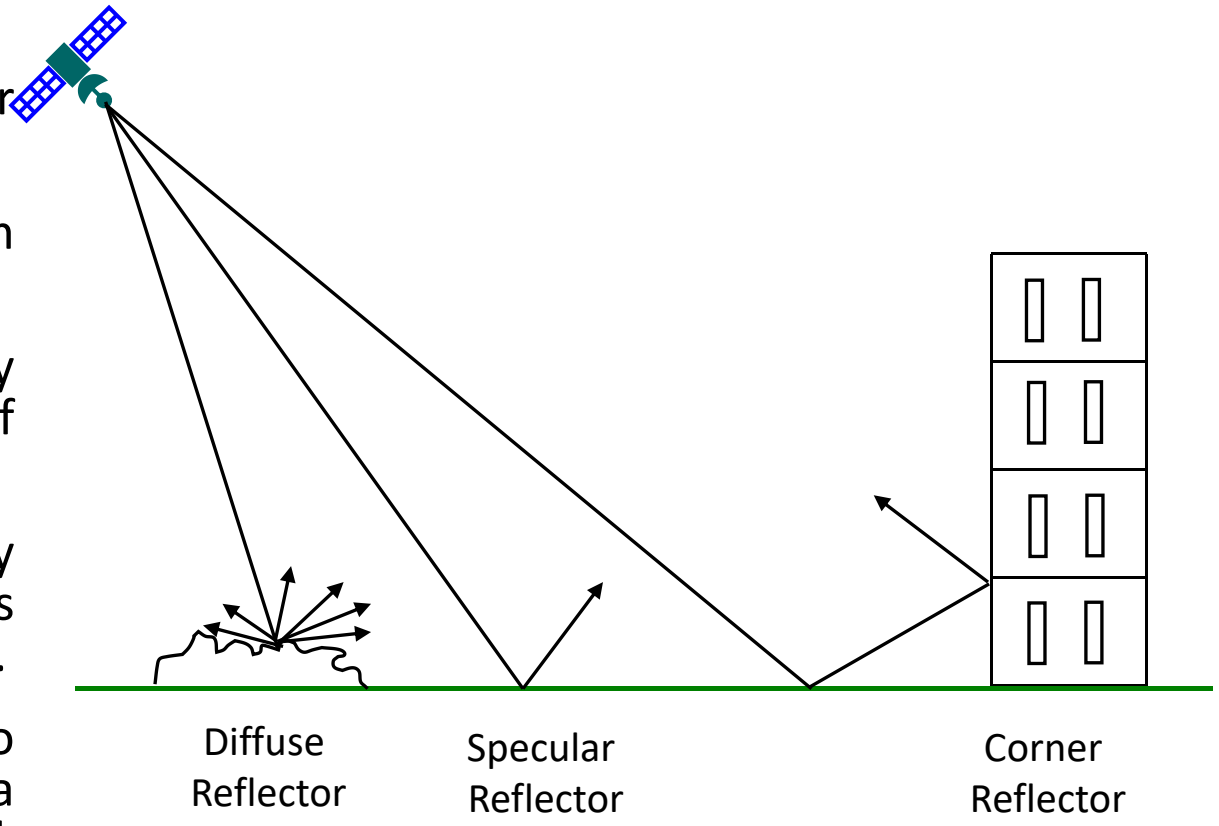
  Water: ~80; Dry surface (rocks & soils): 3 to 8; Metals: 2 to 5;
- Material with high dielectric constant reflects more EM energ
- Amount of soil moisture influences how deep incident EM e
  the material.
- In soils with high moisture content incident energy penetra
  more-brighter return.



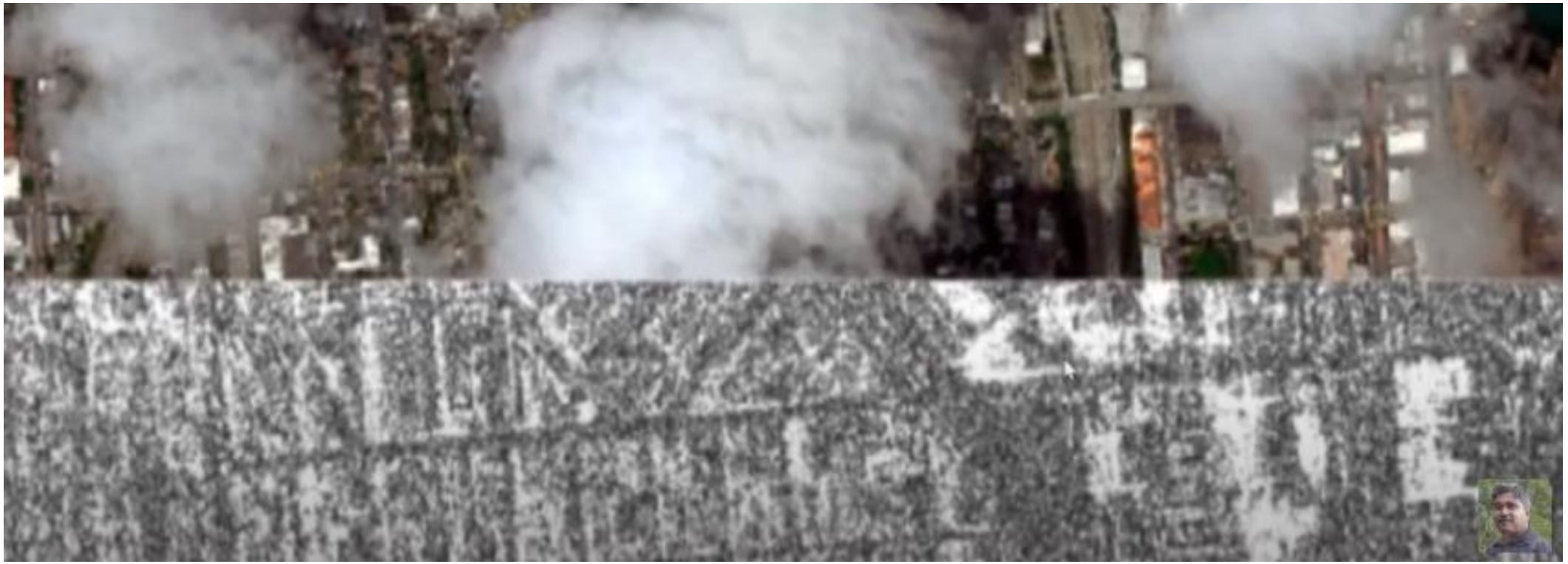


TP: Surface Roughness

- Surface roughness is a terrain property that strongly influences the strength of radar backscatter.
- Surfaces can be either smooth or rough or intermediate.
- Smooth surfaces reflect all the energy away from the antenna and will be resulted in dark tone.
- Rough surfaces diffusely scatter the energy equally in all directions irrespective of the angle of incidence.
- Intermediate surfaces scatter the energy diffusely but not equally since a portion of energy is specularly reflected and the rest diffusely scattered.
- One may use the modified Rayleigh criteria to predict what the earths surface would look like in a radar image if the surface roughness characters and radar system parameters are known.



Visual Interpretation: SAR vs Optical



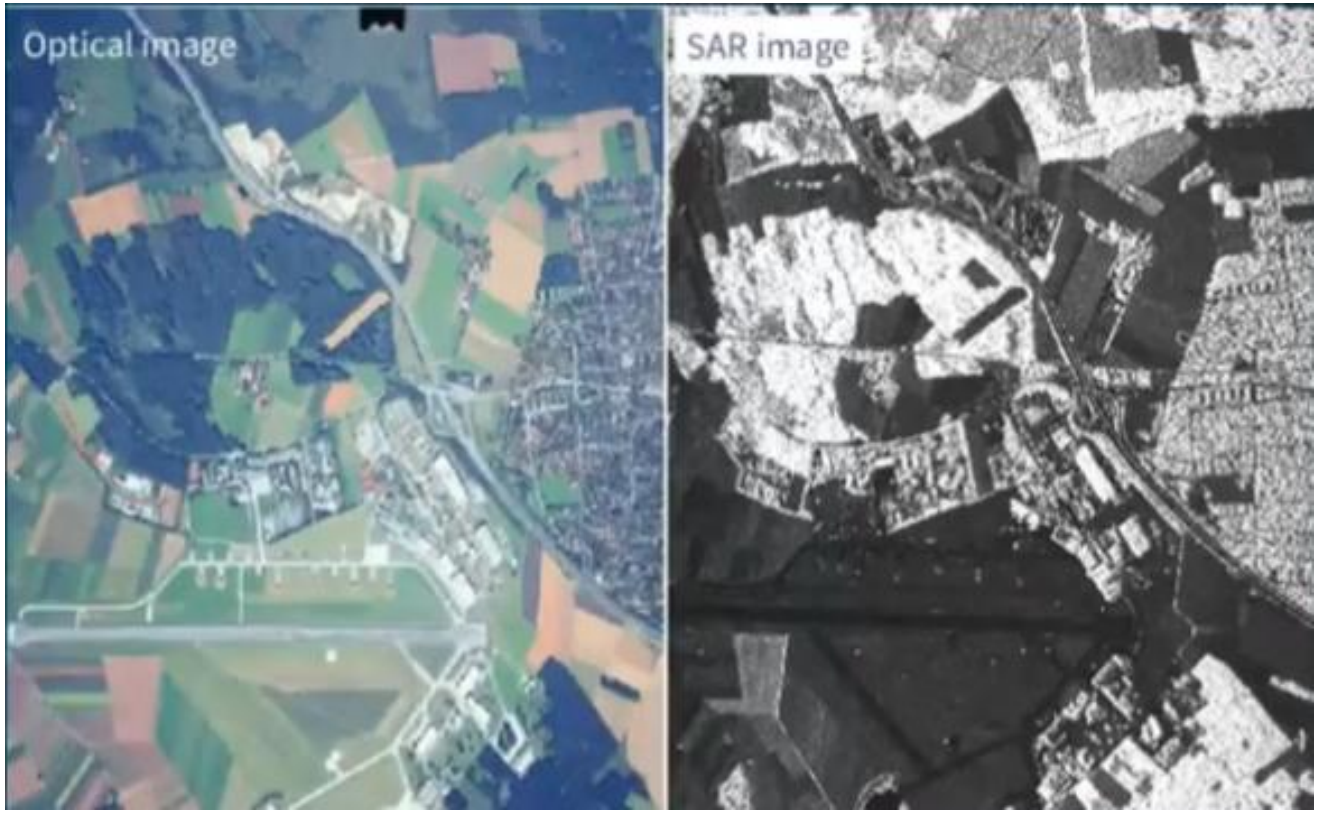
Optical Image

SAR Image

- SAR sensors are sensitive to earth **surface structures** and **moisture content** (dielectric properties)
- Optical sensors are sensitive to **illumination** and **molecular properties** of the earth surface features

Courtesy: Basudeb Bhatta

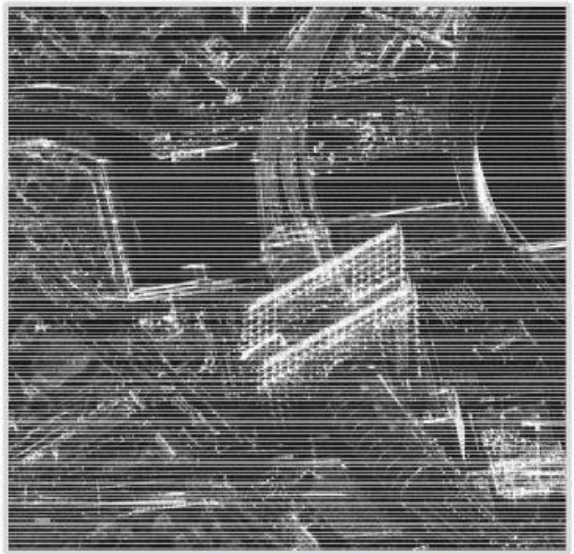
Visual Interpretation: SAR vs Optical



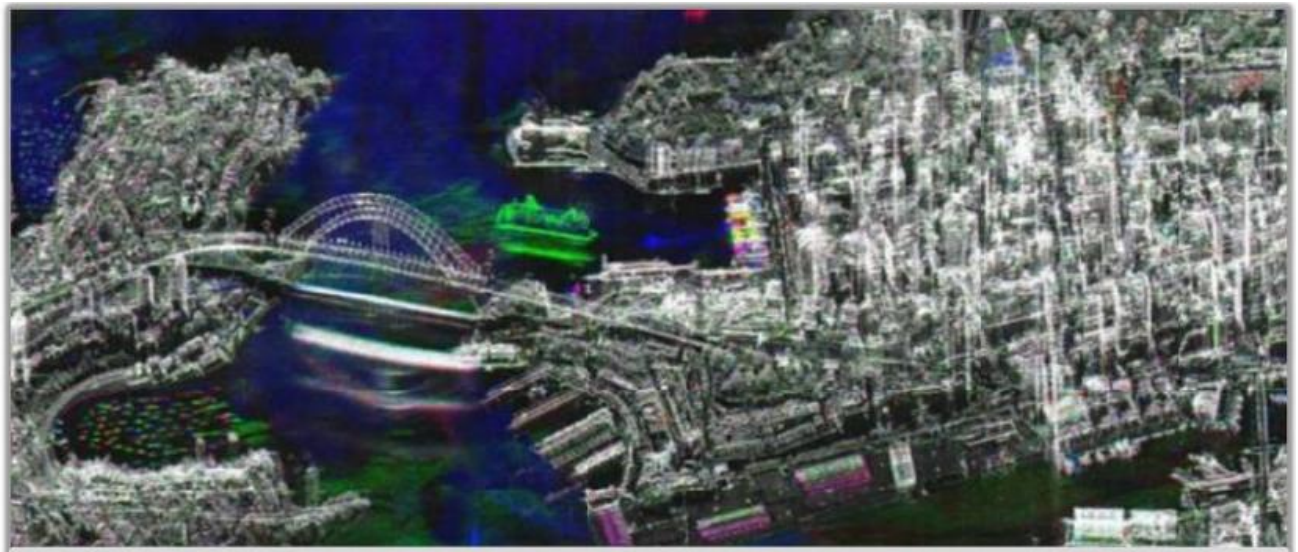


Courtesy: Basudeb Bhatta

Visual Interpretation of SAR Images



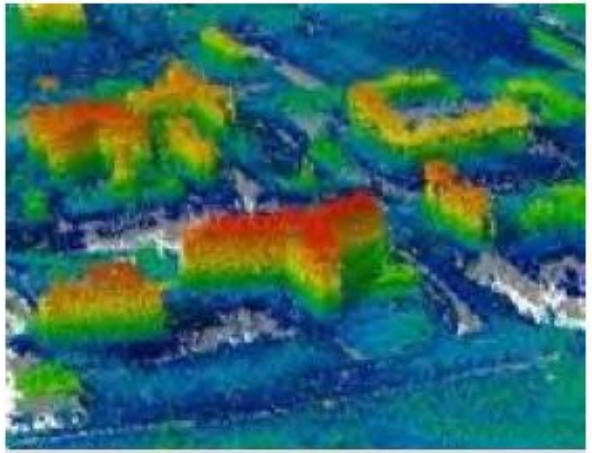
TerraSAR-X, Berlin



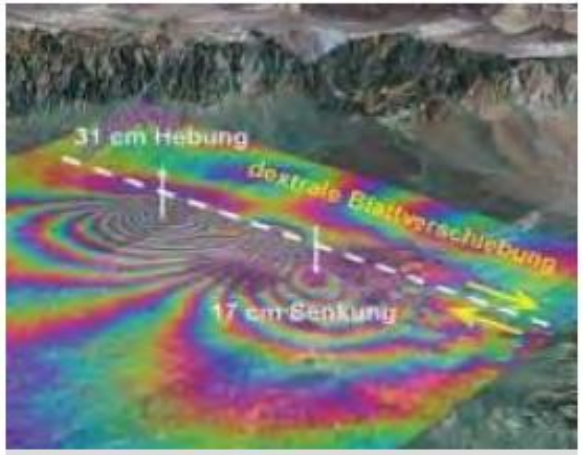
TerraSAR-X, multi-temporal, Sydney, Australia



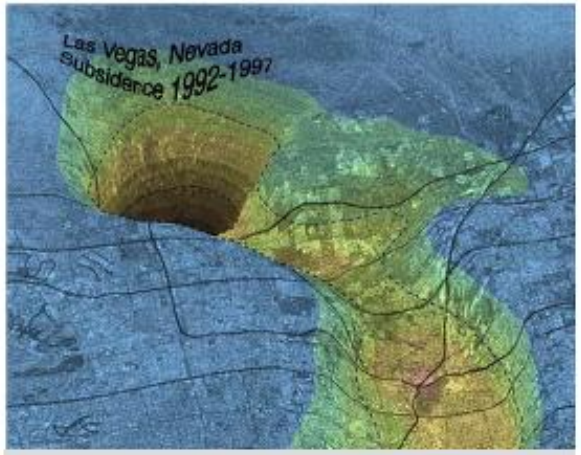
3D Mapping
(Digital Elevation Model)



Tomography
(Urban Mapping)

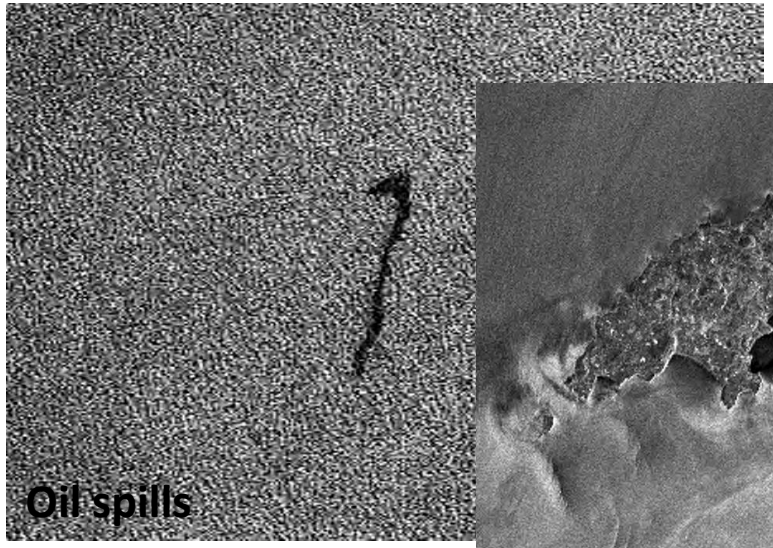


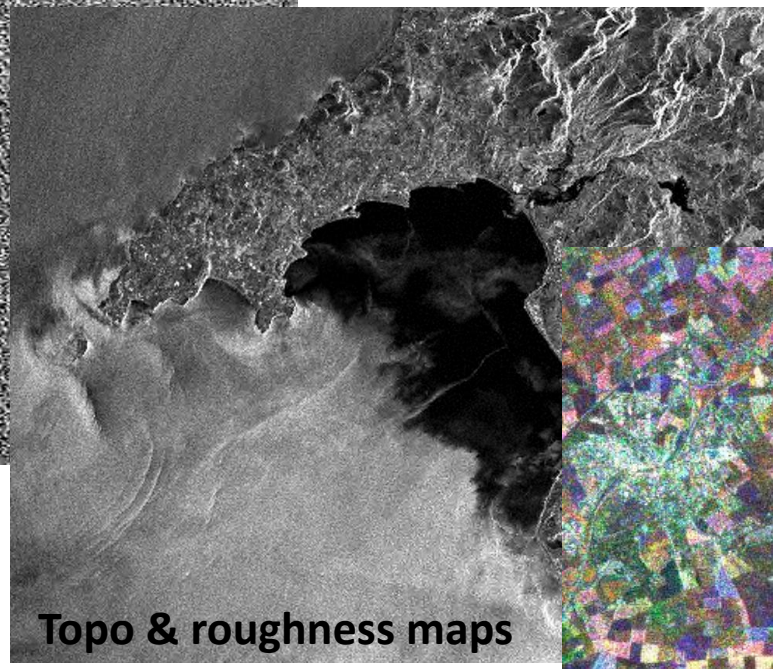
Differential Interferometry
(Earthquake deformation)

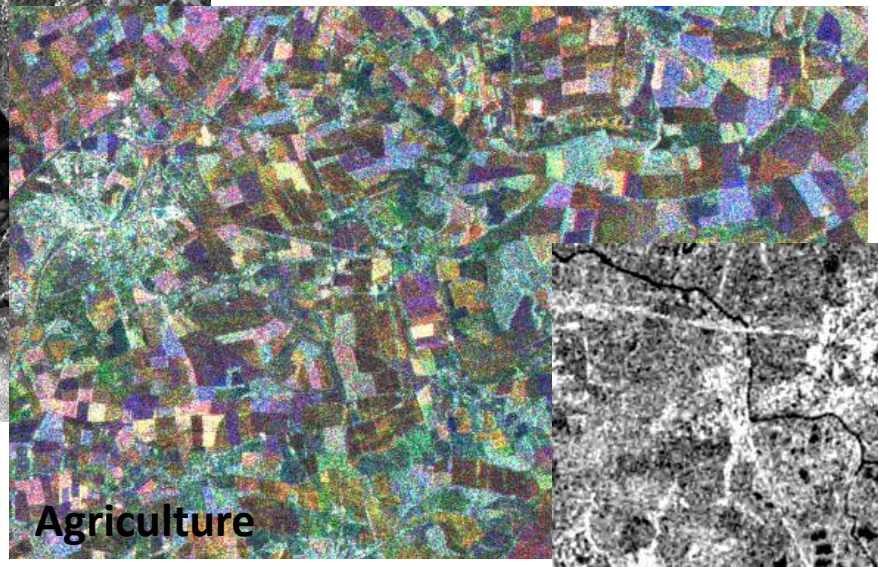


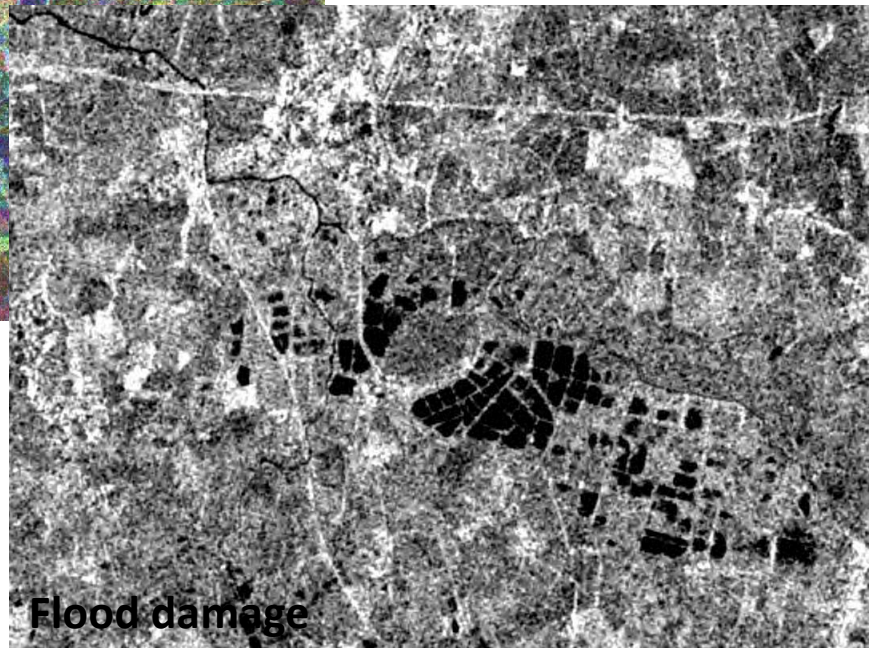
Differential Interferometry
(Subsidence)

Visual Interpretation of SAR Images





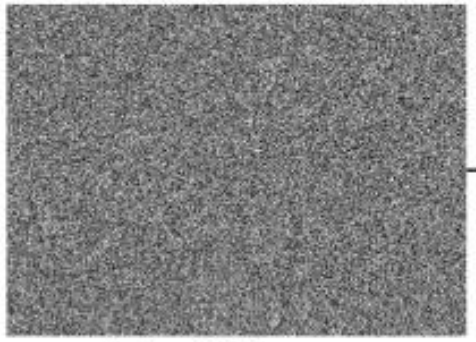


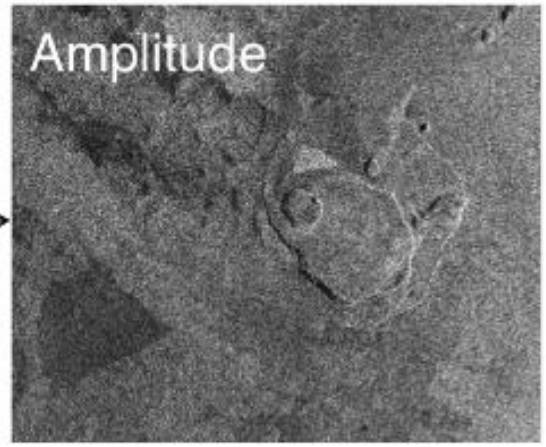


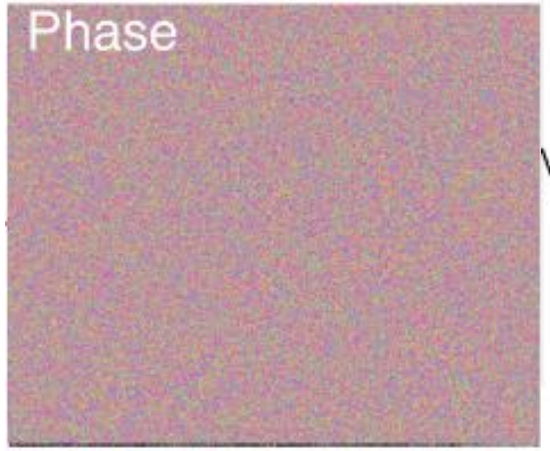
Visual Interpretation of SAR Images

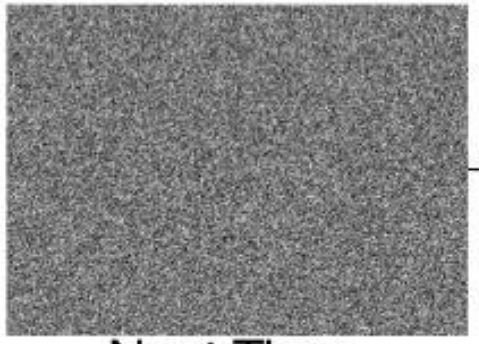


Complex Images (Real & Imaginary parts)

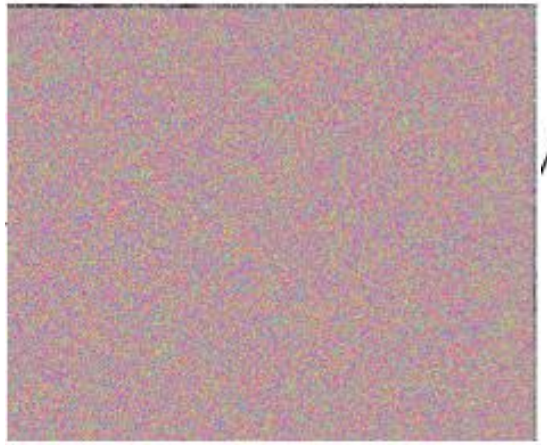




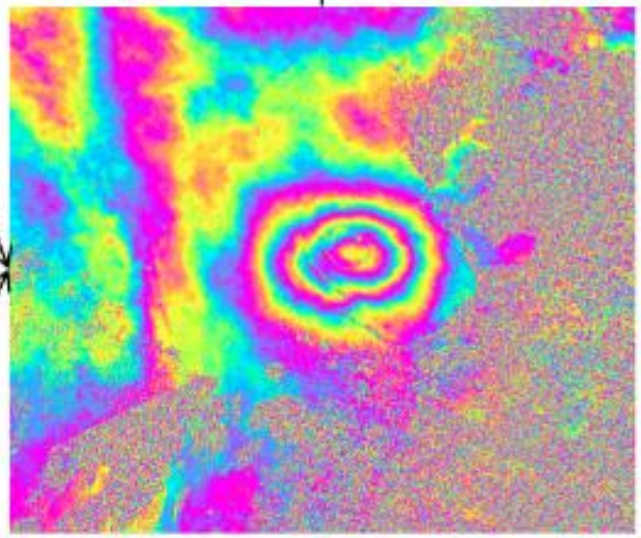








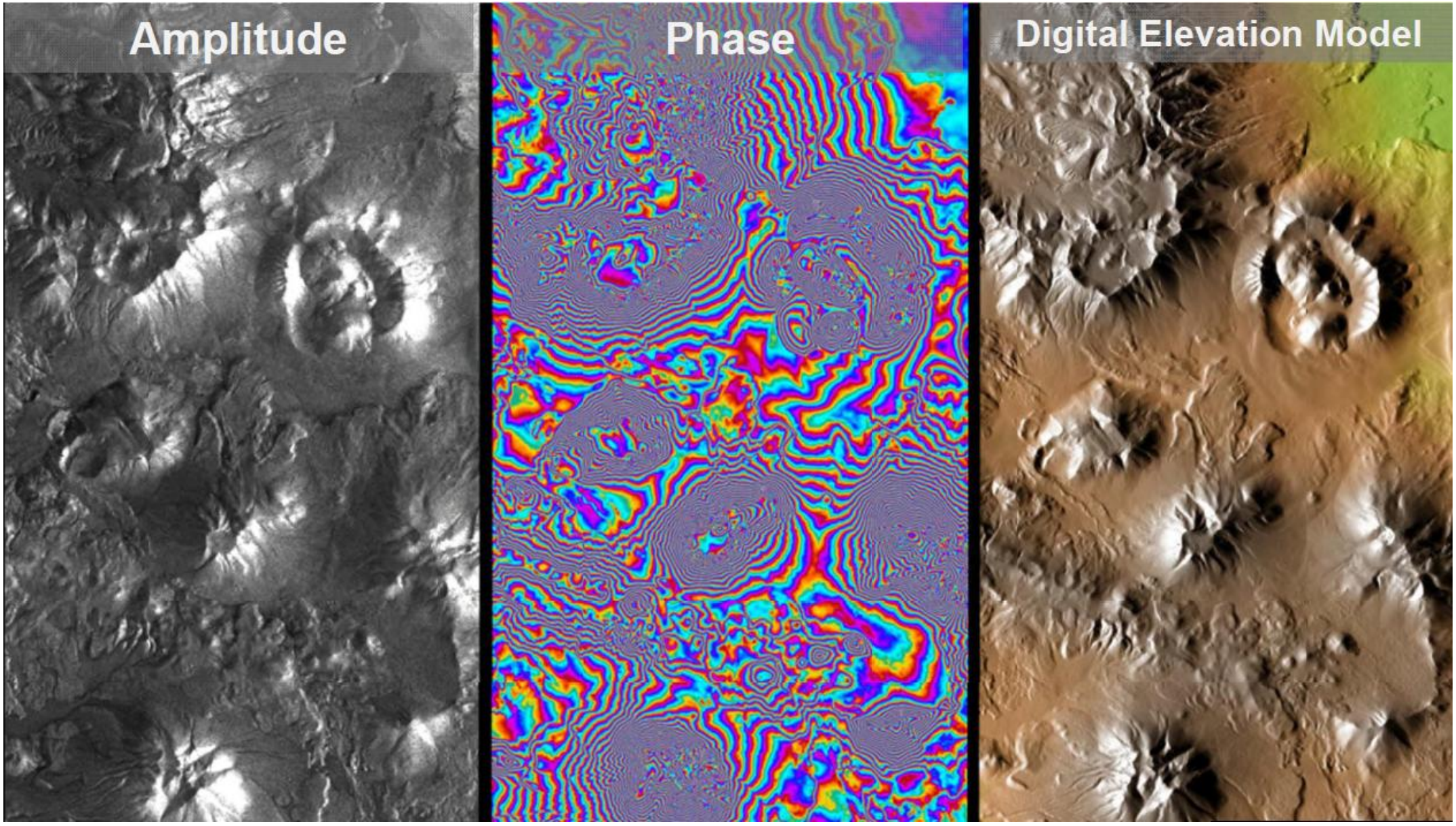
Modeling



Magic of interferometry

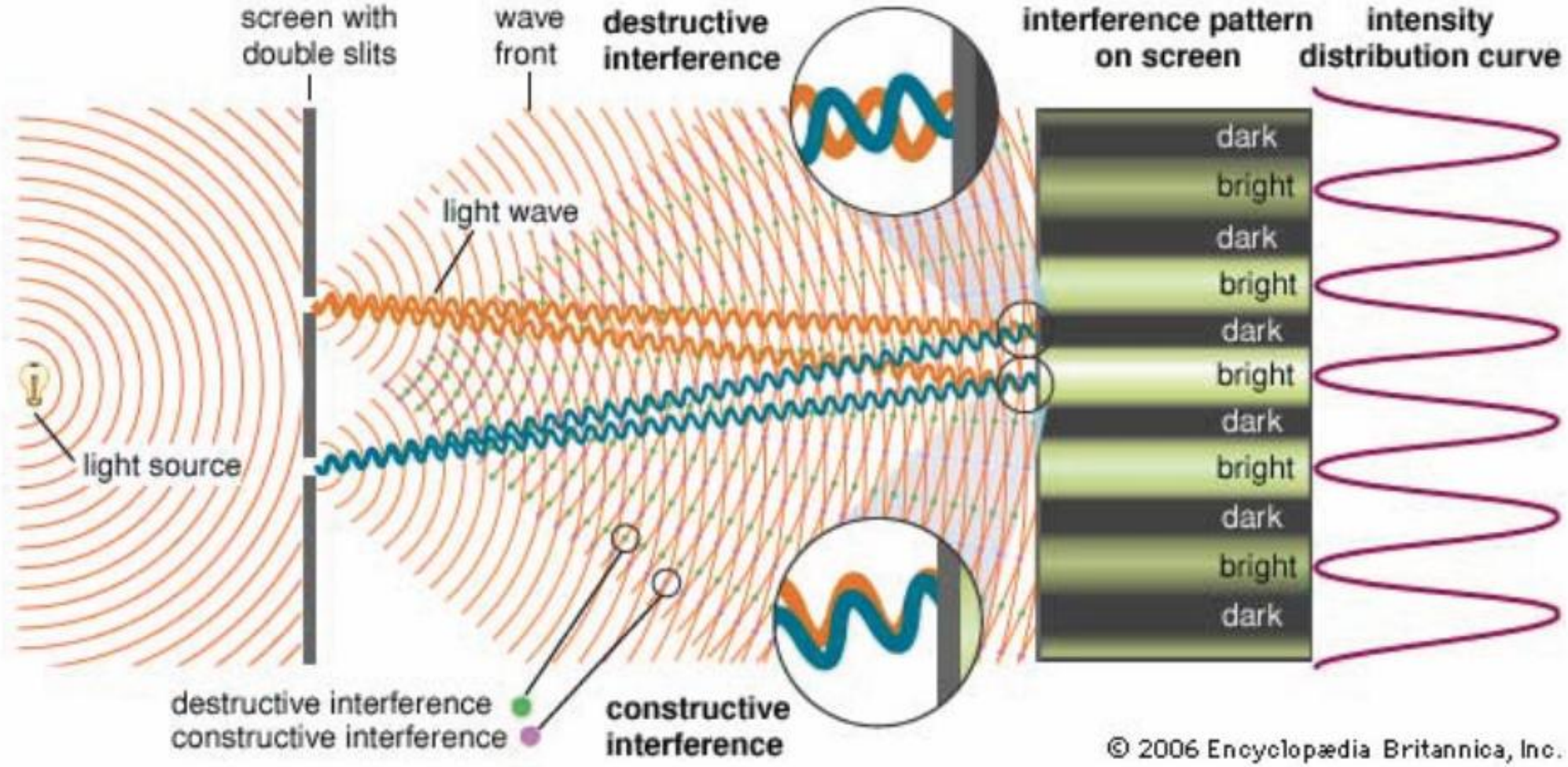
Courtesy: NASA JPL

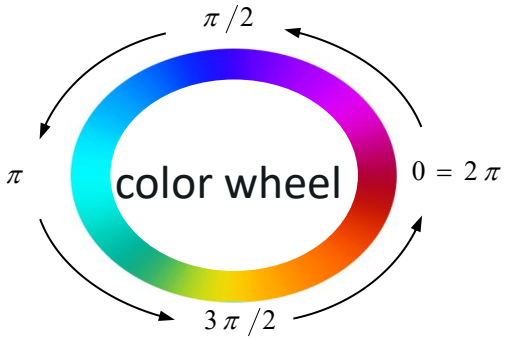
SAR Image of Atacama Desert, Chile

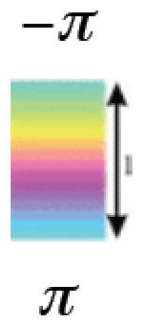


Wave Interference

Young's Double Slit Experiment (Interferometer)

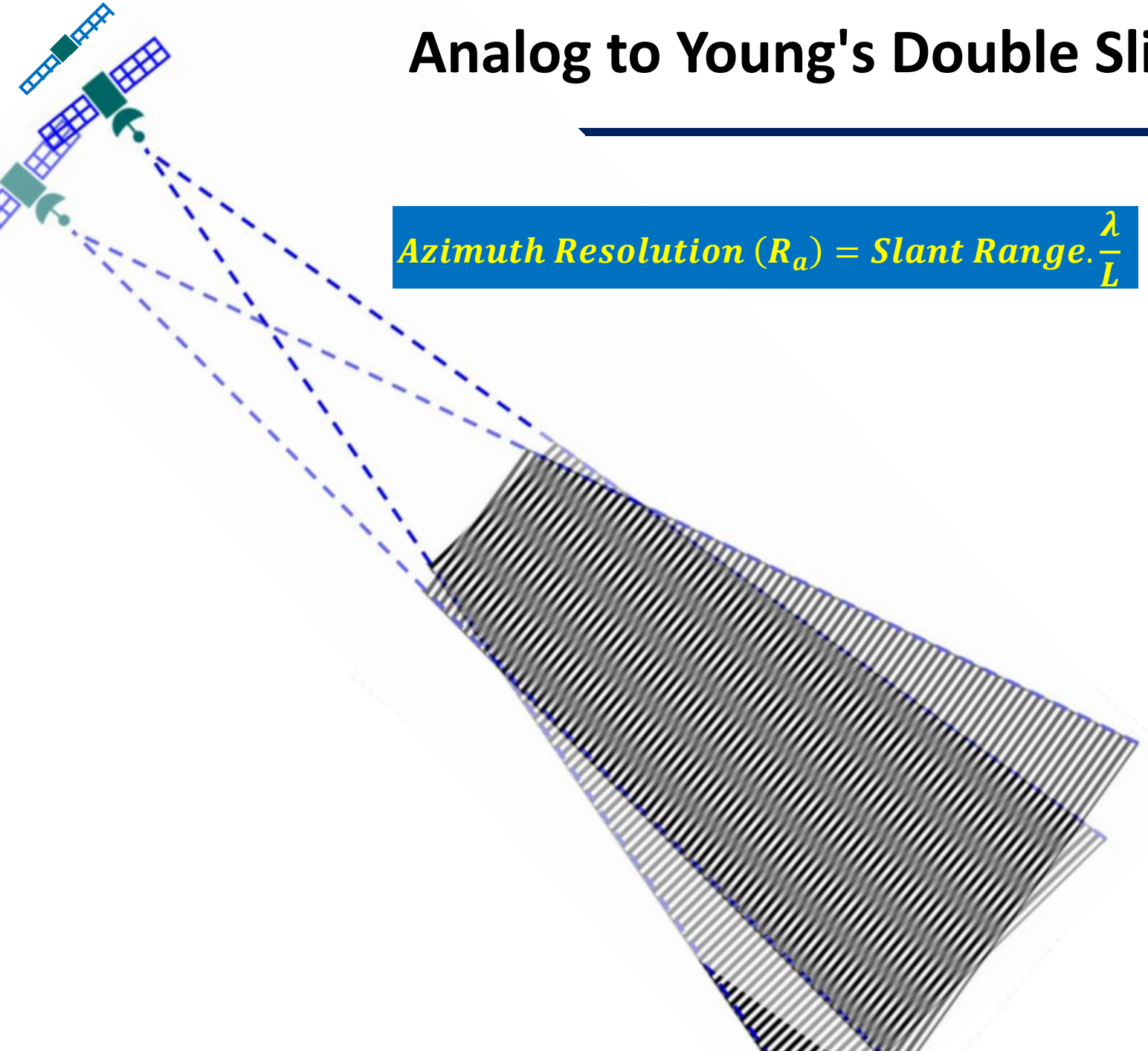


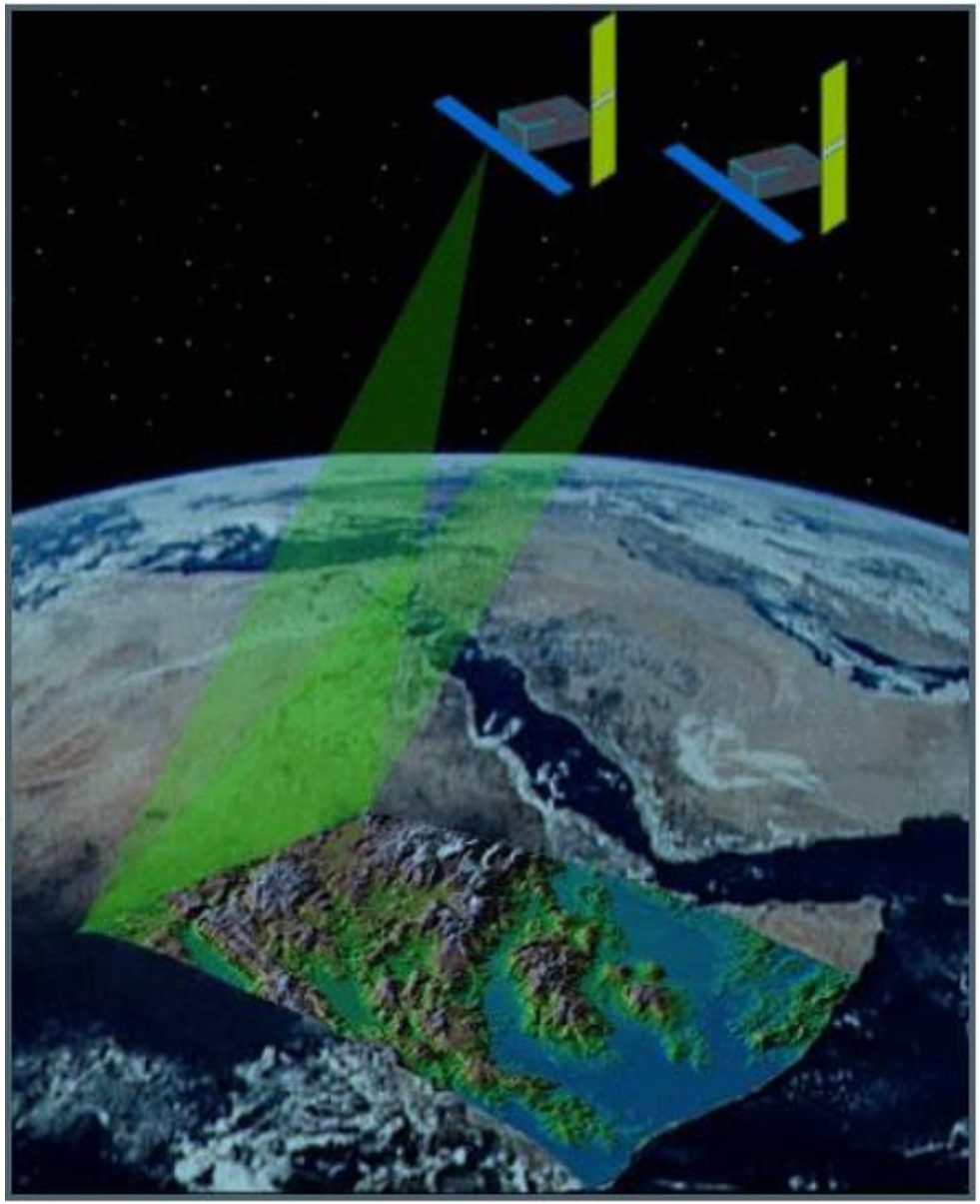




Analog to Young's Double Slit Experiment

$$Azimuth\ Resolution\ (R_a) = Slant\ Range.\frac{\lambda}{L}$$





Mathematical Formulation of InSAR

Free-space wave equation, derived from Maxwell's EM Equations:

$$\nabla^2\vec{E} + \frac{\omega^2}{c^2}\vec{E} = 0$$

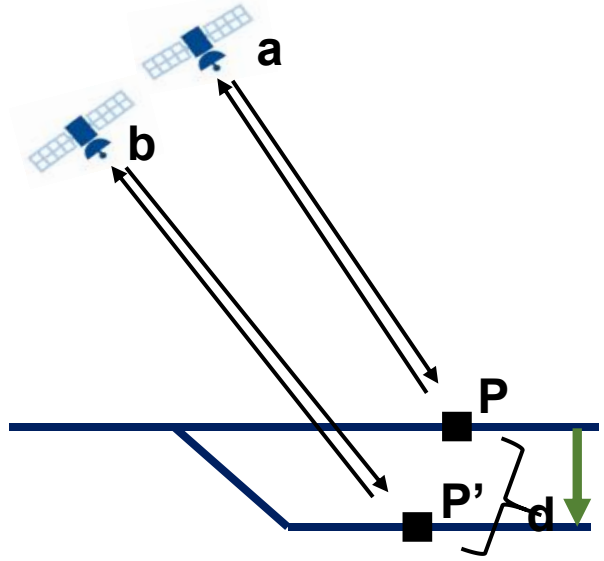
$$\vec{E} = E_0 e^{i\psi} = E_0 e^{i(\vec{k}.\vec{r} - \omega t + \phi)}$$

$$\phi_a = \phi_{geom-a} + \phi_{scat-a}; \qquad \phi_b = \phi_{geom-b} + \phi_{scat-b}$$

$$\vec{E} = \vec{E}_a * \vec{E}_b{}^{\dagger} = E_a . E_b . \, e^{i(\phi_b - \phi_a)}$$

$$\Delta\phi_{int} = \phi_b(t_b, x_b, r_b) - \phi_a(t_a, x_a, r_a) = \phi_{geom-b} - \phi_{geom-a}$$

$$\Delta\phi_{int} = \frac{4\pi}{\lambda}(bP' - aP) = \frac{4\pi}{\lambda}(bP' - bP) + \frac{4\pi}{\lambda}(bP - aP)$$



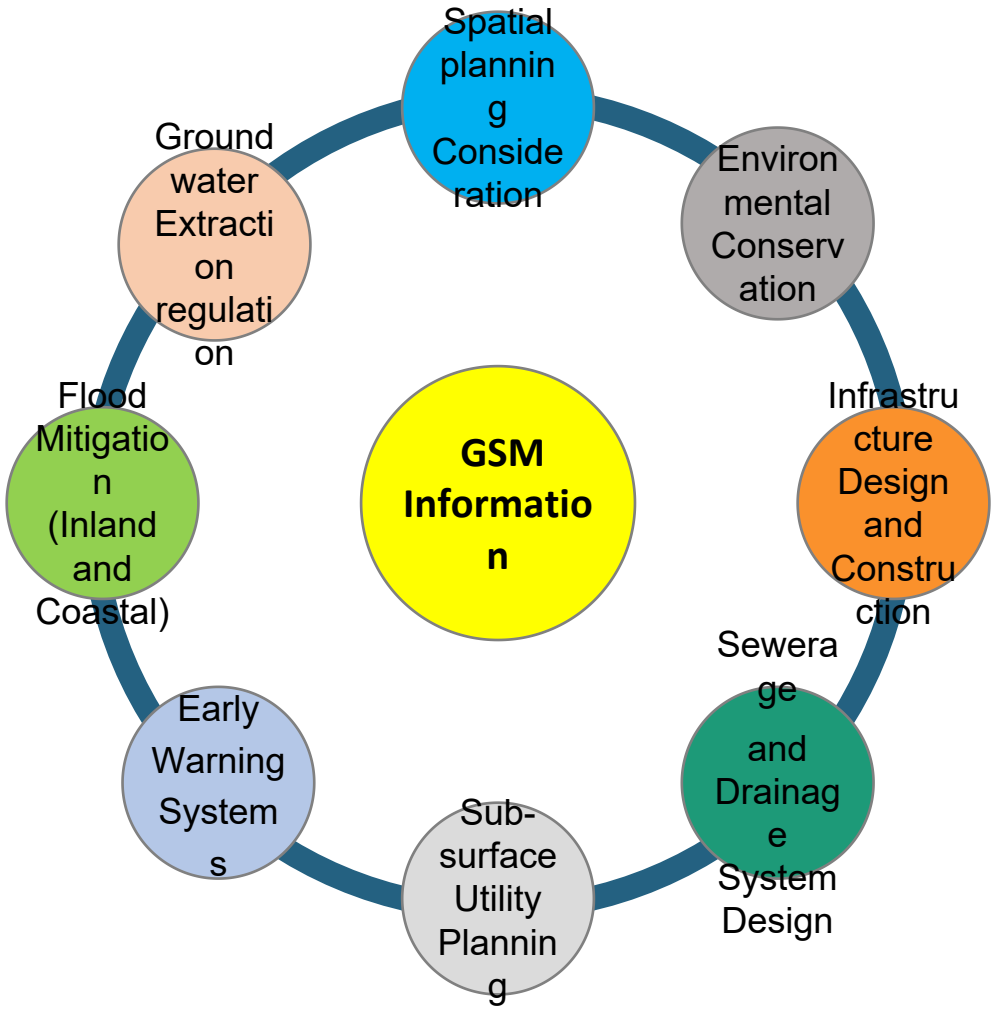
$$\Delta\phi_{D-int} = \Delta\phi_{int} - \phi_{topo_simu} = \phi_{disp}$$

$$\Delta\phi_{int} = \phi_{disp} + \phi_{set} + \phi_{topo_res} + \phi_{flat} + \phi_{atm} + \phi_{orb} + \phi_{noise} + 2\pi n$$

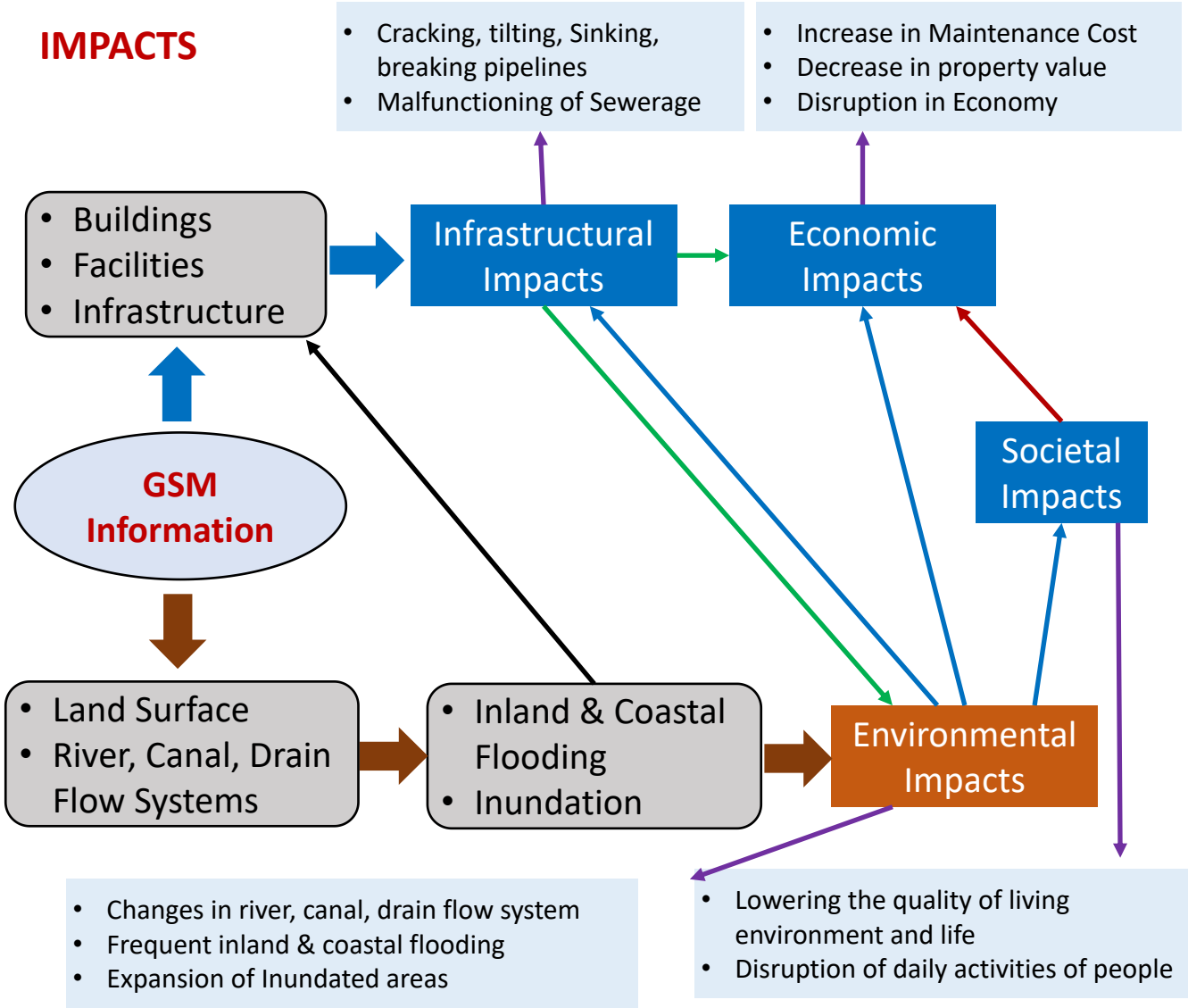
$$\phi_{disp} = +\frac{4\pi}{\lambda}\text{Disp.}(PP')$$

Ground & Infrastructural Deformation

IMPORTANCE



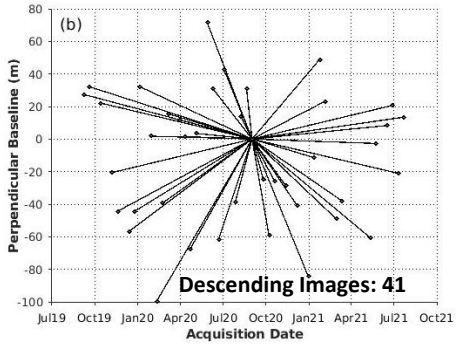
IMPACTS

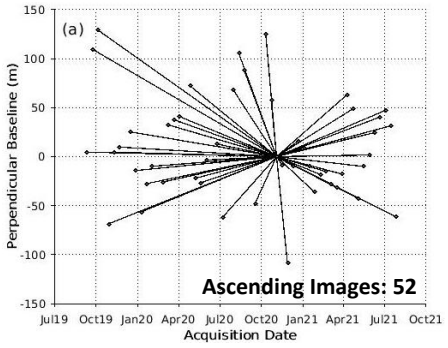


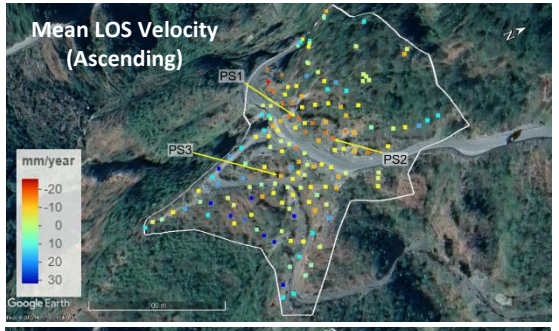
Land Surface Movement in Sirmaur District (H.P.)

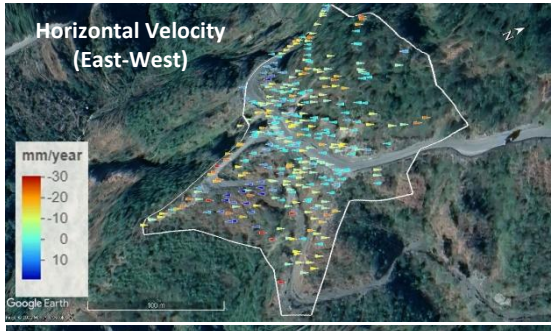
Highlights

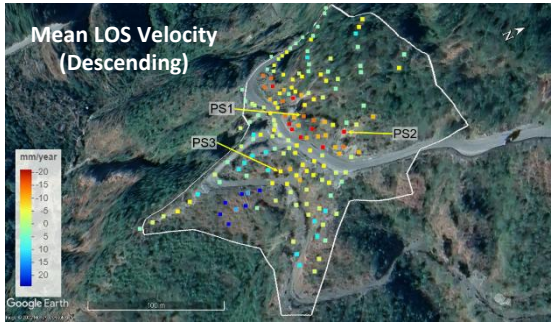
- Pre-failure data (Sep 2019-Jul 2021) was used for temporal prediction
- ***Inverse Velocity Methods Used***
- **Slope instability** started from Jul 2020 (~1 year before the actual event)

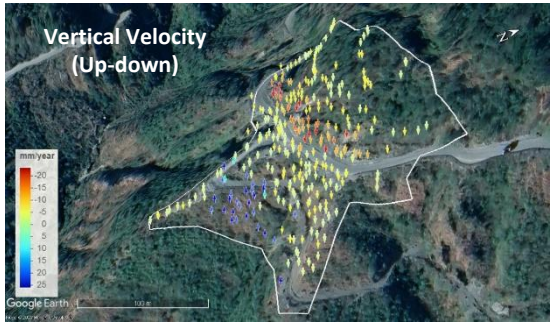


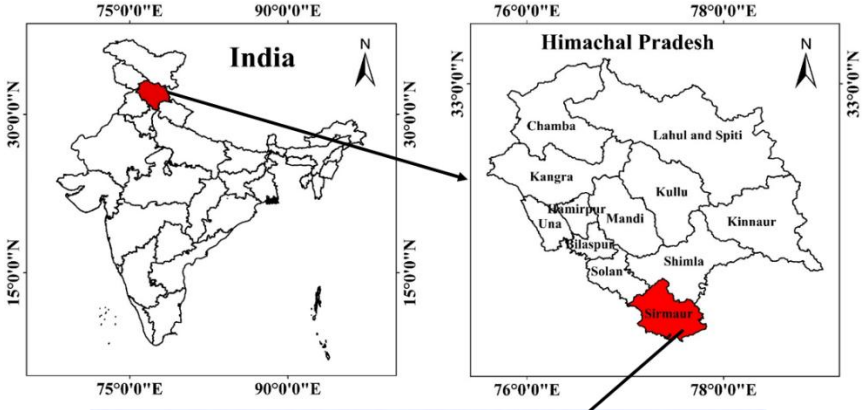












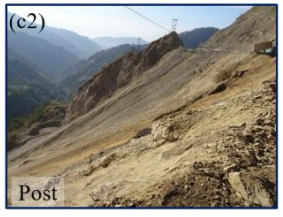


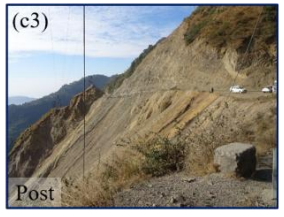
Landslide Location





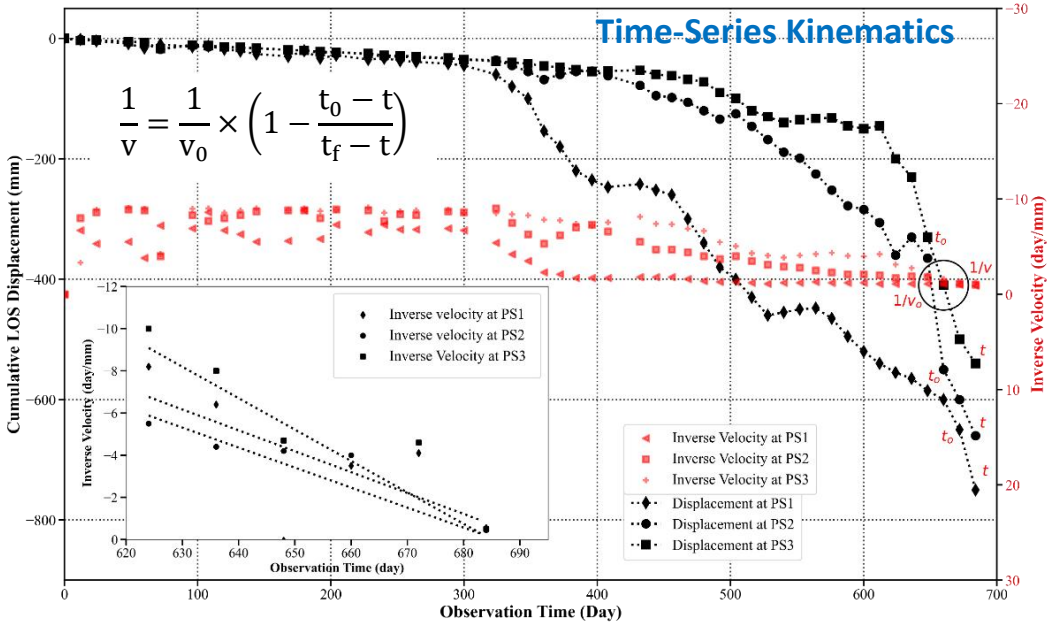






Time-Series Kinematics

$$\frac{1}{v} = \frac{1}{v_0} \times \left(1 - \frac{t_0 - t}{t_f - t}\right)$$

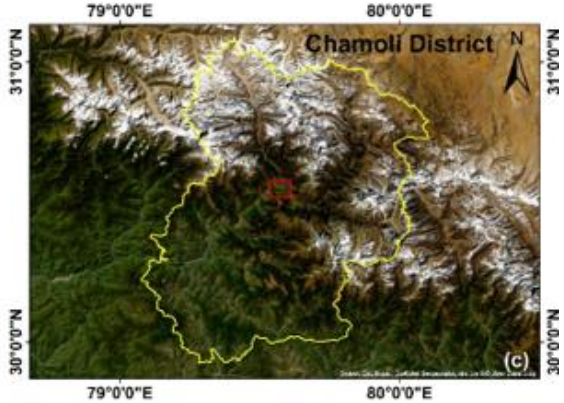


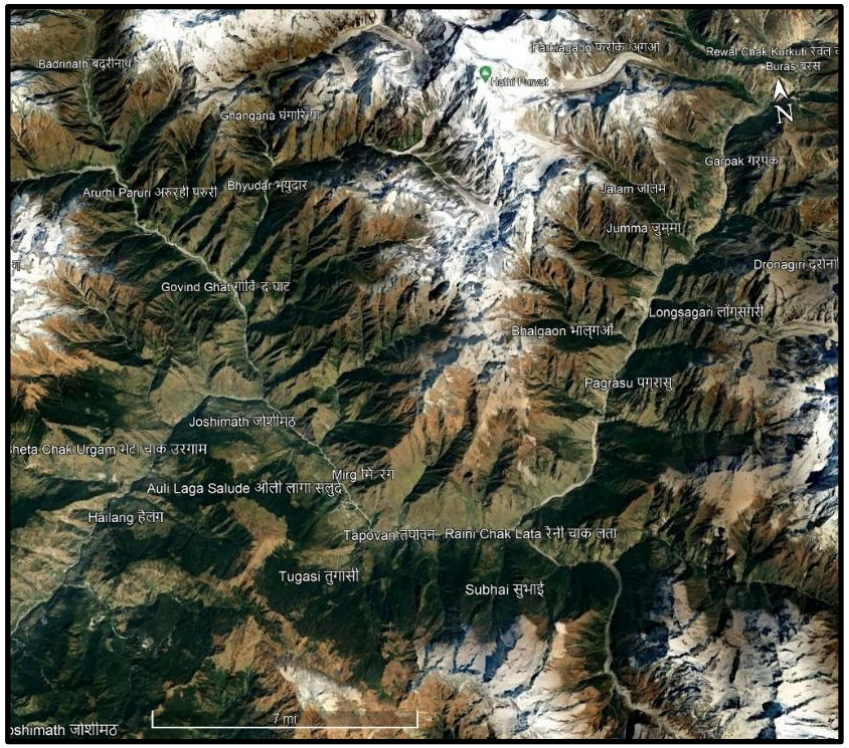
Assessment of Surface Movement of Joshimath Town



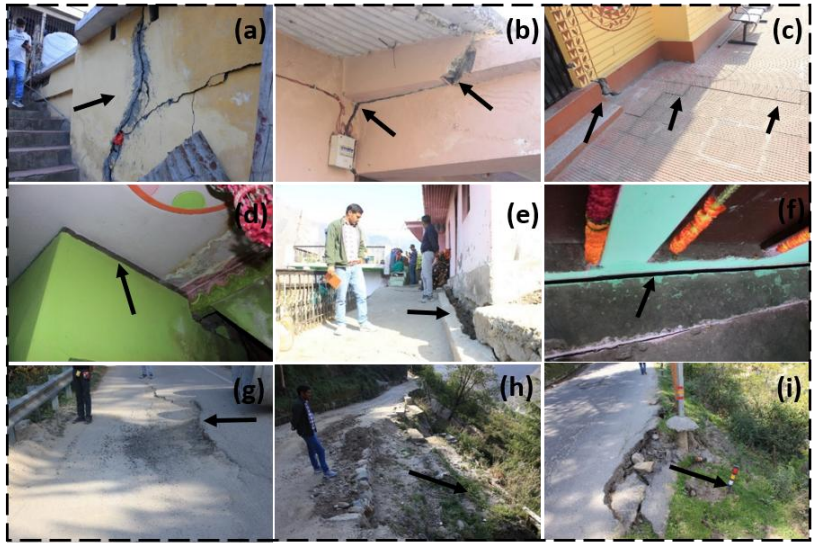








April 2022



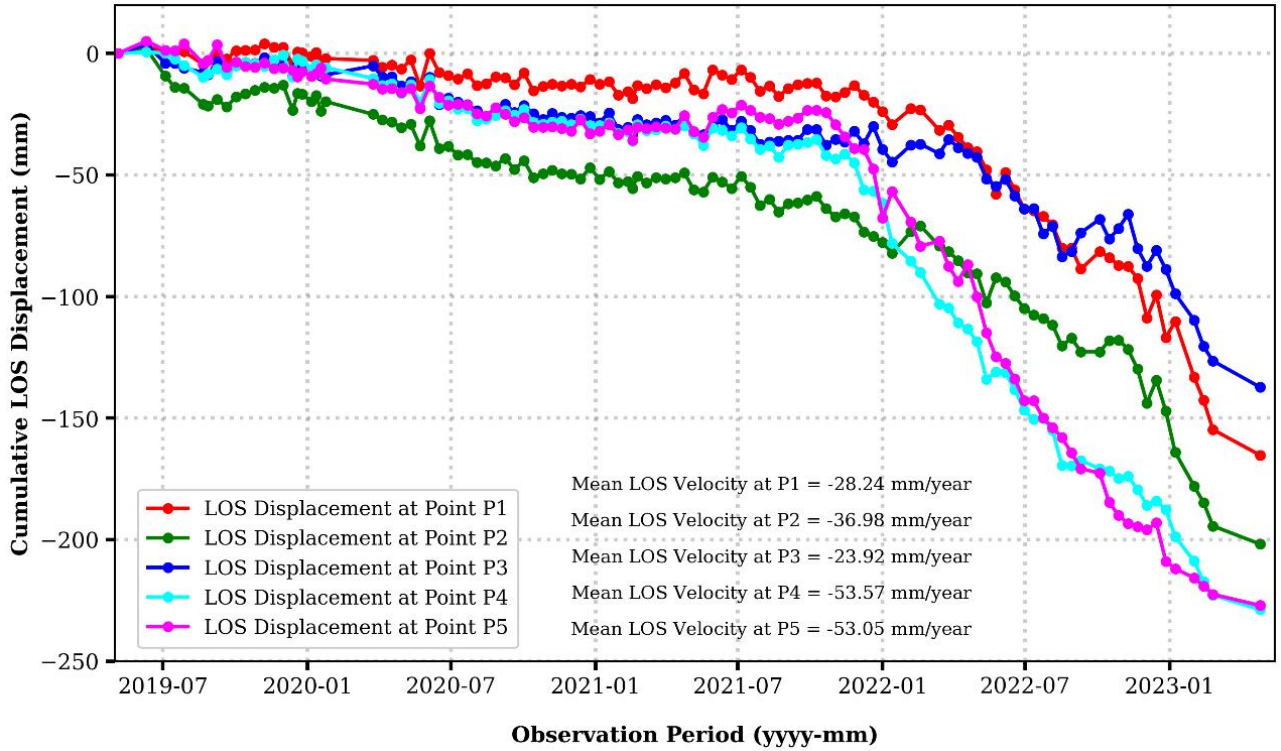
January 2023

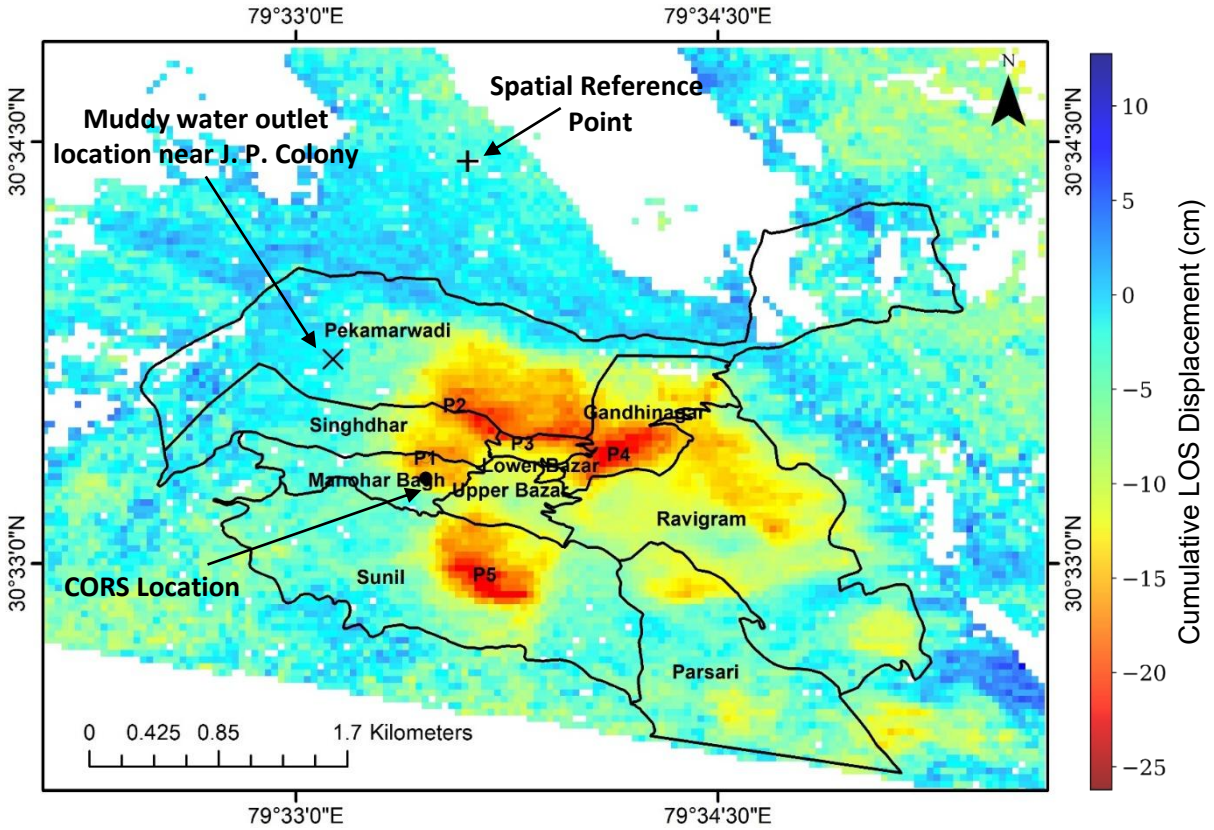


March 2023

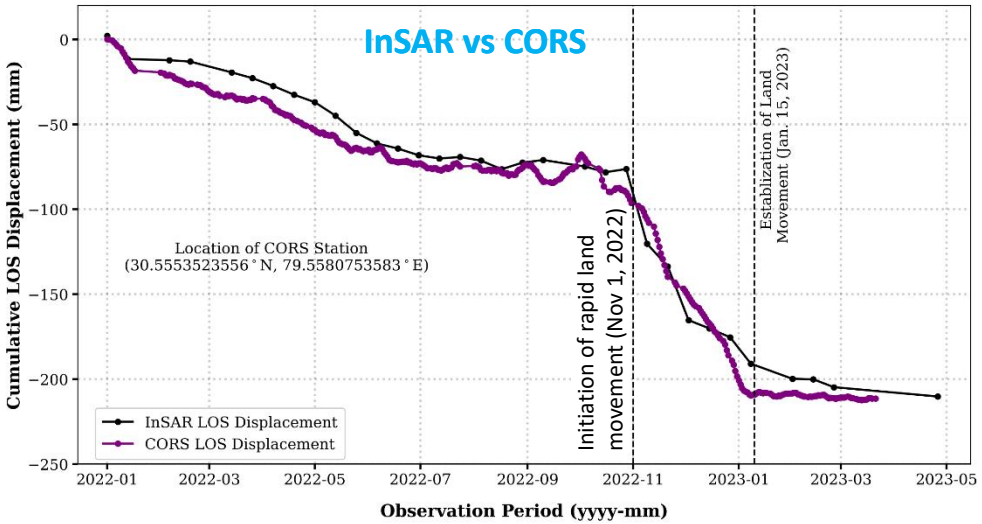


Assessment of Surface Movement of Joshimath Town



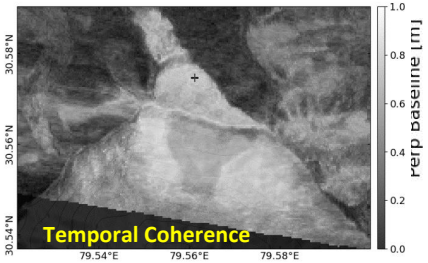


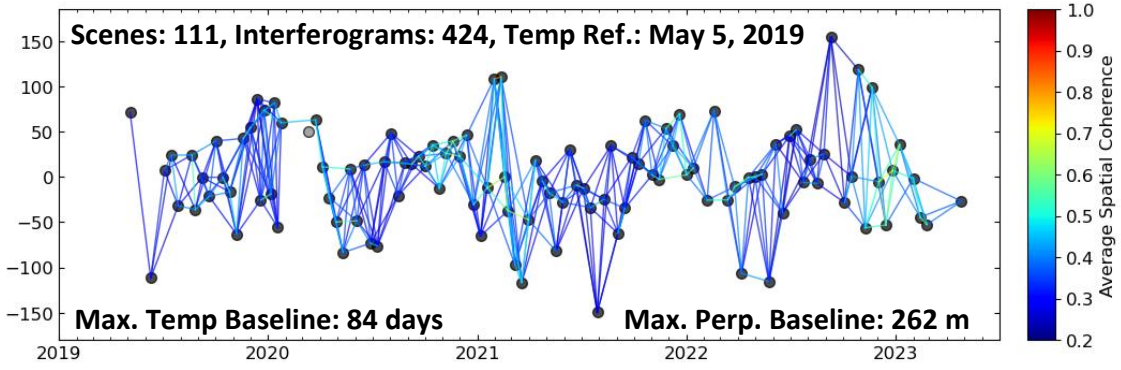
Investigation Period: May 5, 2019 – April 26, 2023



Highlights

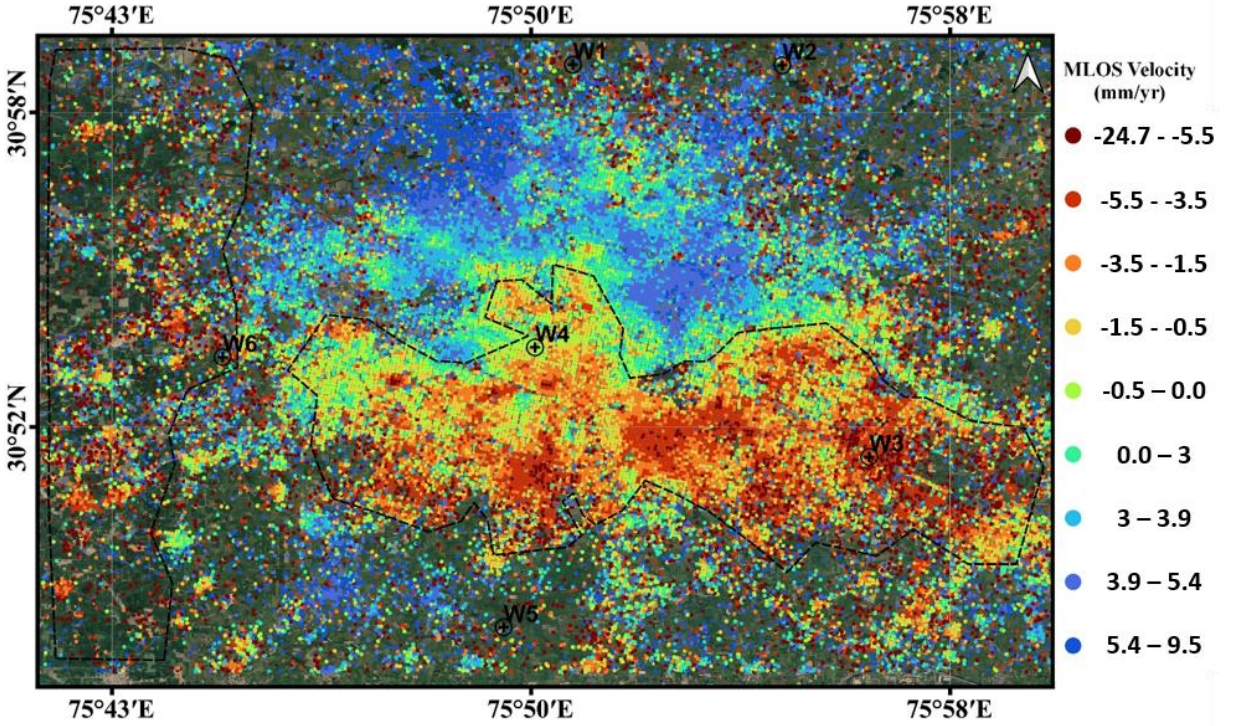
- Applied SBAS Technique
- Velocity & acceleration calculated
- Validated with CORS observation

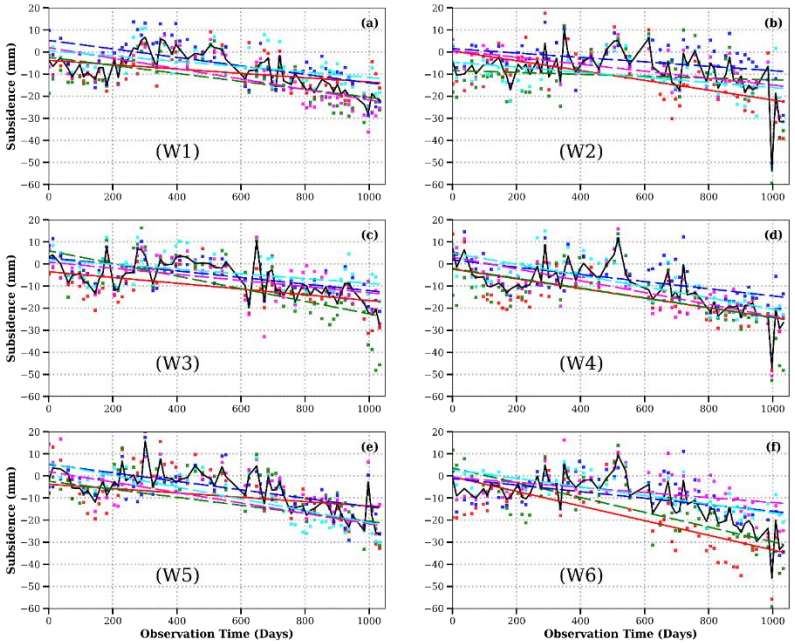


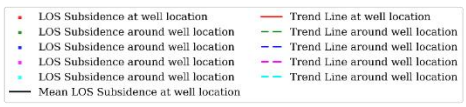


Investigation of Land Subsidence of Ludhiana City, Punjab



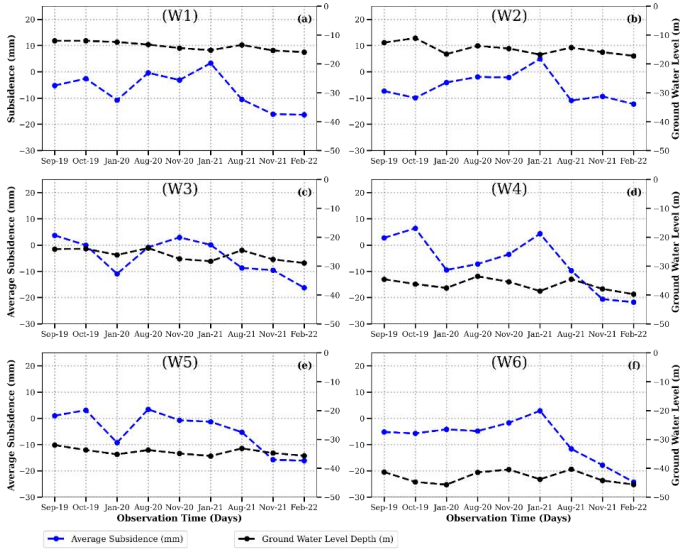


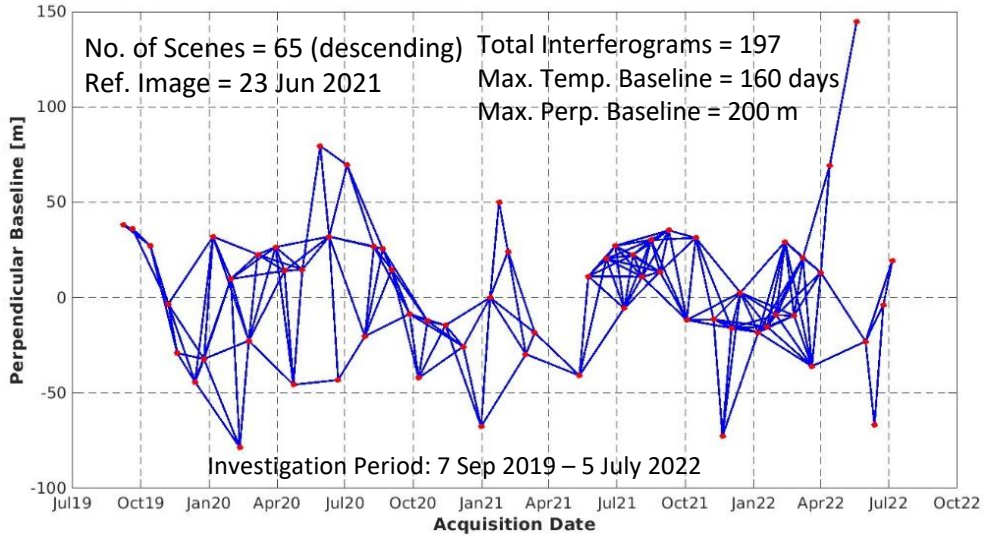




Highlights

- SBAS results compared with si GWL wells observations between.
- Southern Part of city sufferin from subsidence





Monitor tunneling activities: ground settlement due to volume loss

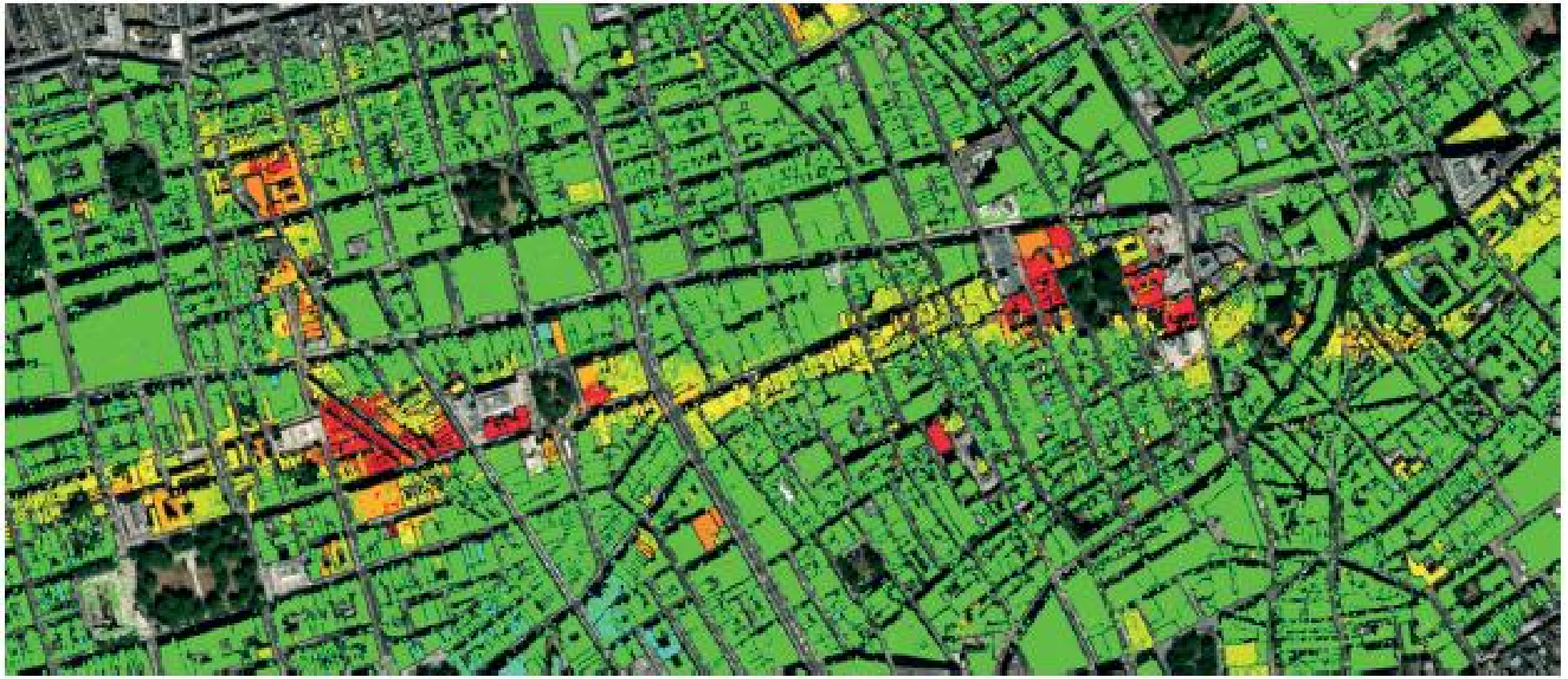
- InSAR monitoring is particularly suited to monitor ground surface displacement resulting from tunneling activities and to measure its impact on existing buildings and infrastructures.
- During tunneling advancement, InSAR is used to monitor ground settlement due to volume loss, verify surveying reference points and map any widening of the settlement outside the calculated ZOI.



https://www.sixense-group.com/en/services/monitoring-testing/satellite-monitoring/atlas-insar-monitoring

Urban Monitoring: Buildings

- **Monitor** the **impact** of your construction activities on adjacent buildings and infrastructure
- Assess the **structural health** of **buildings** in zones vulnerable to subsidence, differential settlement, exoansive soils etc.
- Obtain **unambiguous evidence** in case of unjustified claims

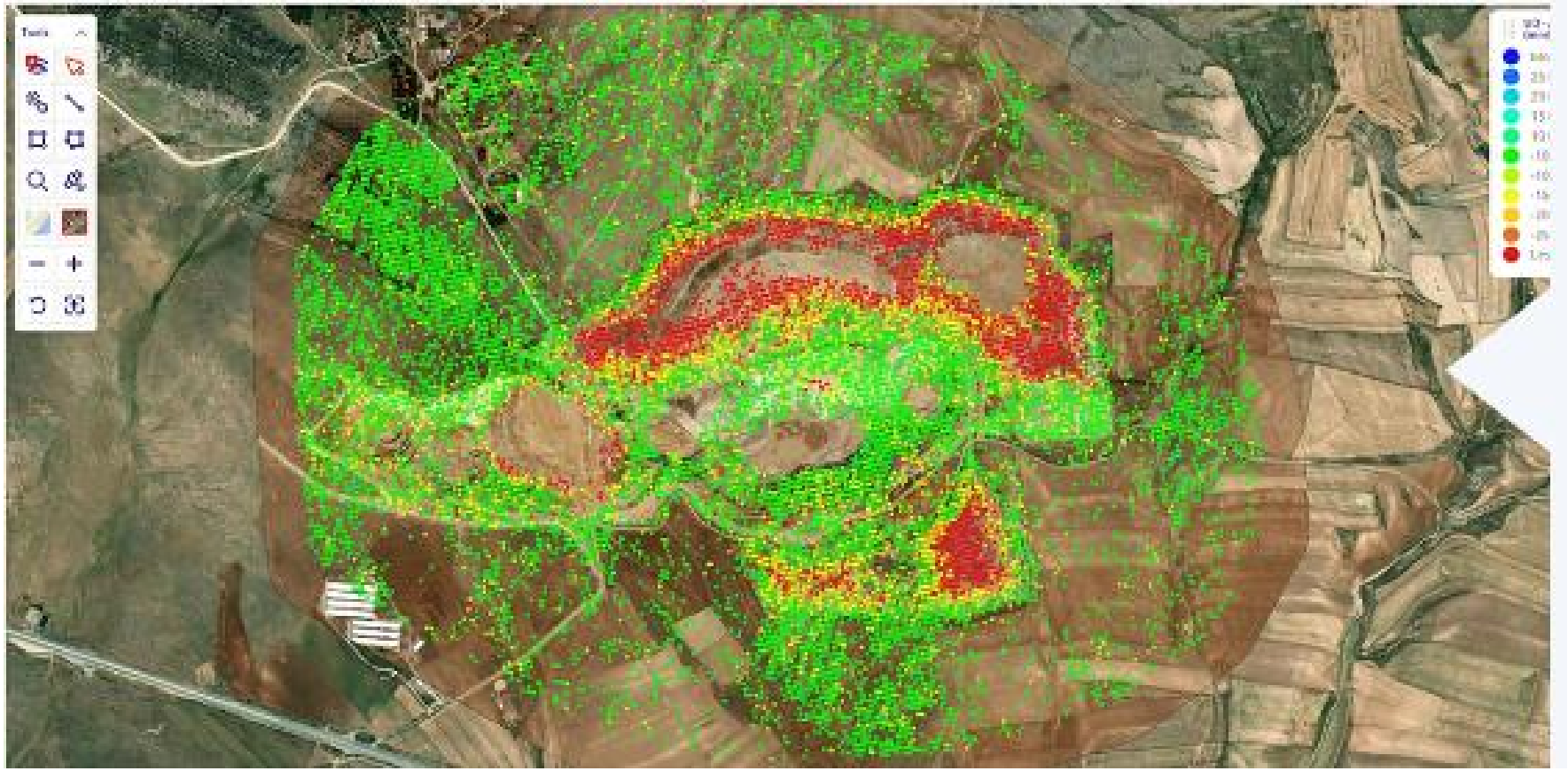


Impact of Tunneling works on nearby buildings. Risk index for single building is provided. Images TerraSAR-X@ Airbus

Source: https://www.sixense-group.com/en/services/monitoring-testing/satellite-monitoring/atlas-insar-monitoring

Risk reduction in mining and tailings dams

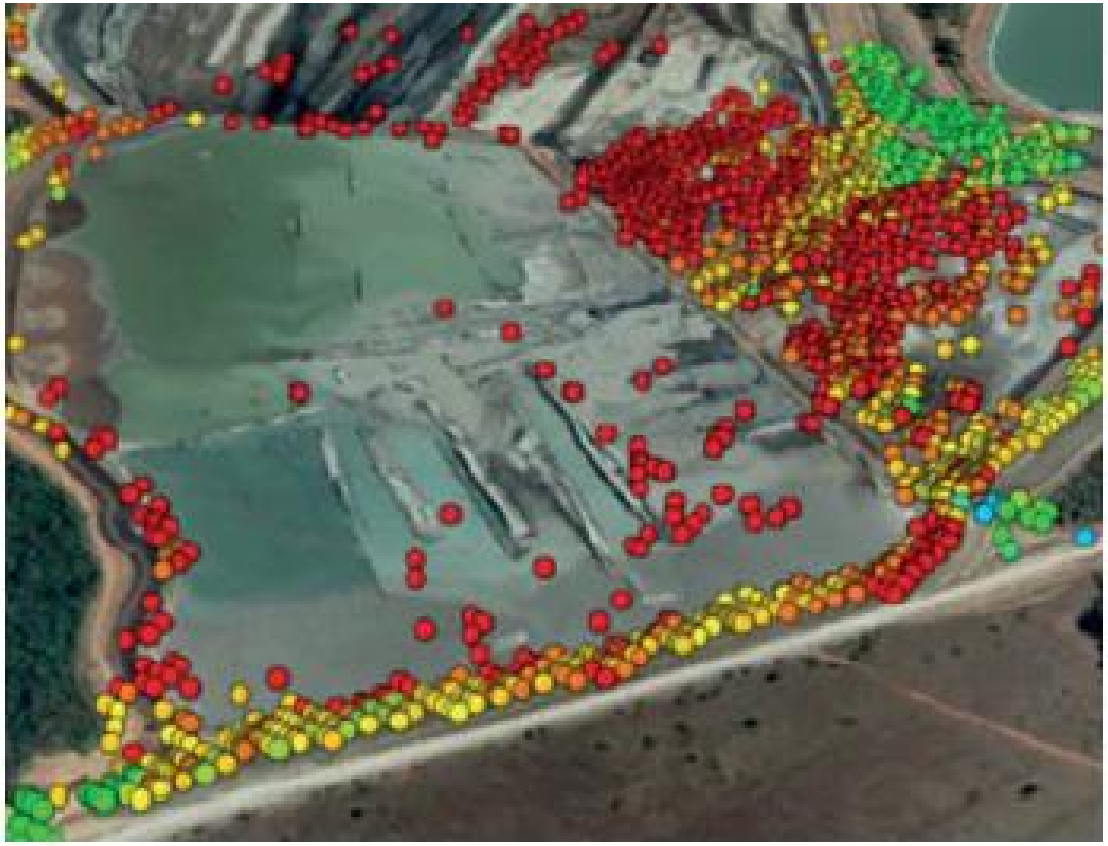
- InSAR Monitoring enables customized monitoring programs to support mining activities.
- It can track ground motion at both underground and open-pit mining sites and can be used to ensure the stability of critical infrastructures, including tailings dams, waste dumps, access roads, and pipelines.
- InSAR helps mining operators identify potential risks early, optimize safety measures, and reduce operational and environmental impacts.

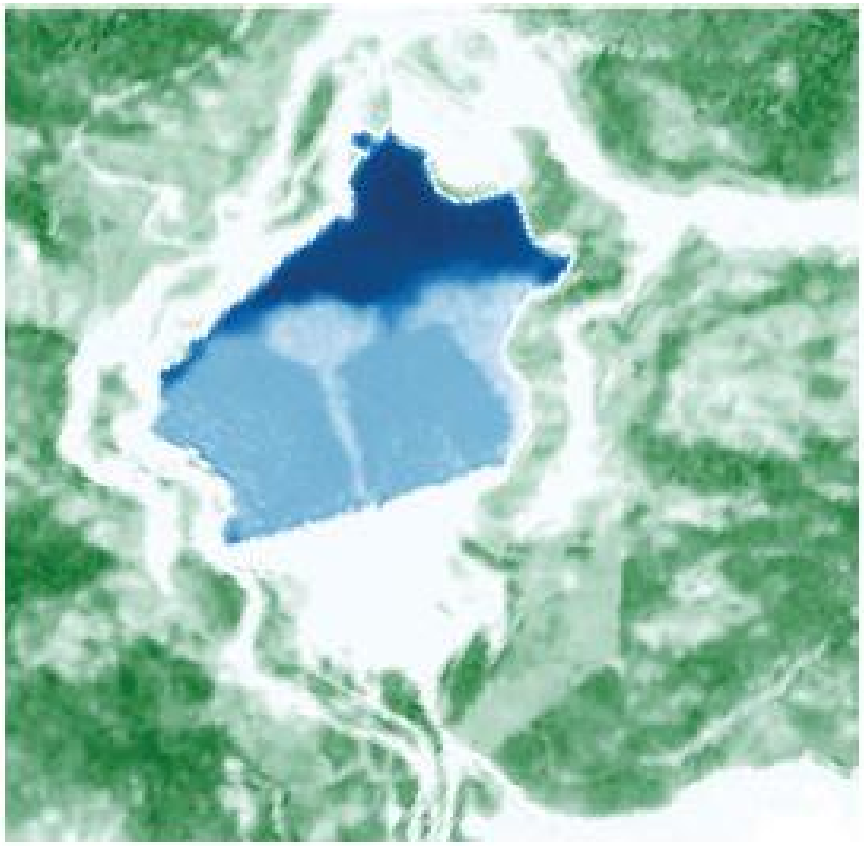


Source: https://www.sixense-group.com/en/services/monitoring-testing/satellite-monitoring/atlas-insar-monitoring

Tailings Storage Facility

- **Understanding** the **Behavior** of the infrastructure over time
- **Period measurements** to **help infrastructure management** and **identify structural weaknesses**
- **Prevention** of **breakage** with detection of precursor movements
- **Mapping** of **surface moisture; Monitoring** of **inactive dams and slopes**





Source: https://www.sixense-group.com/en/services/monitoring-testing/satellite-monitoring/atlas-insar-monitoring

Monitor the impact of construction works

- InSAR monitoring can be applied during the construction phase of any project, providing millions of measurement points with mm precision.
- The area covered by InSAR analyses can extend beyond the defined Zone of Influence (ZOI), making it a valuable tool for detecting deformations that exceed the expected perimeter, such as those resulting from dewatering activities.
- This makes InSAR a potential solution for addressing subsidence claims and complementing in-situ instrumentation.



Source: https://www.sixense-group.com/en/services/monitoring-testing/satellite-monitoring/atlas-insar-monitoring

Early Warning Tool

Detect precursor motion to landslide and slope failures

- InSAR monitoring is an effective tool for detecting precursor movements leading to landslides or slope failures across large areas, such as highways, railways, and pipelines, as well as in remote or critical infrastructure, including mines, tailings dams, and bridges.

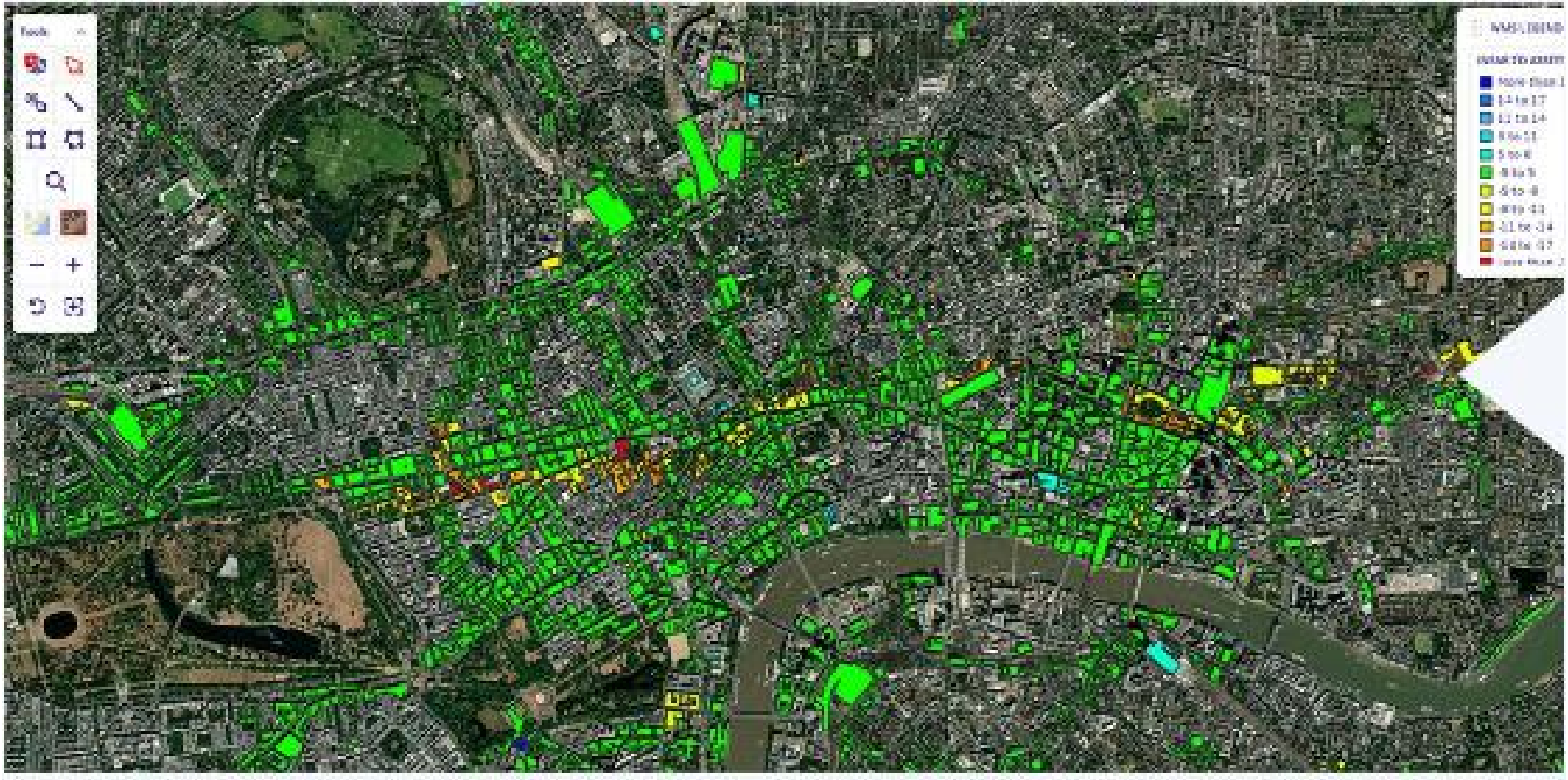
- Non-linear movements or acceleration in motion can be identified, enabling the implementation of early-stage mitigation measures to reduce risks and enhance safety.



Source: https://www.sixense-group.com/en/services/monitoring-testing/satellite-monitoring/atlas-insar-monitoring

Long-Term Maintenance

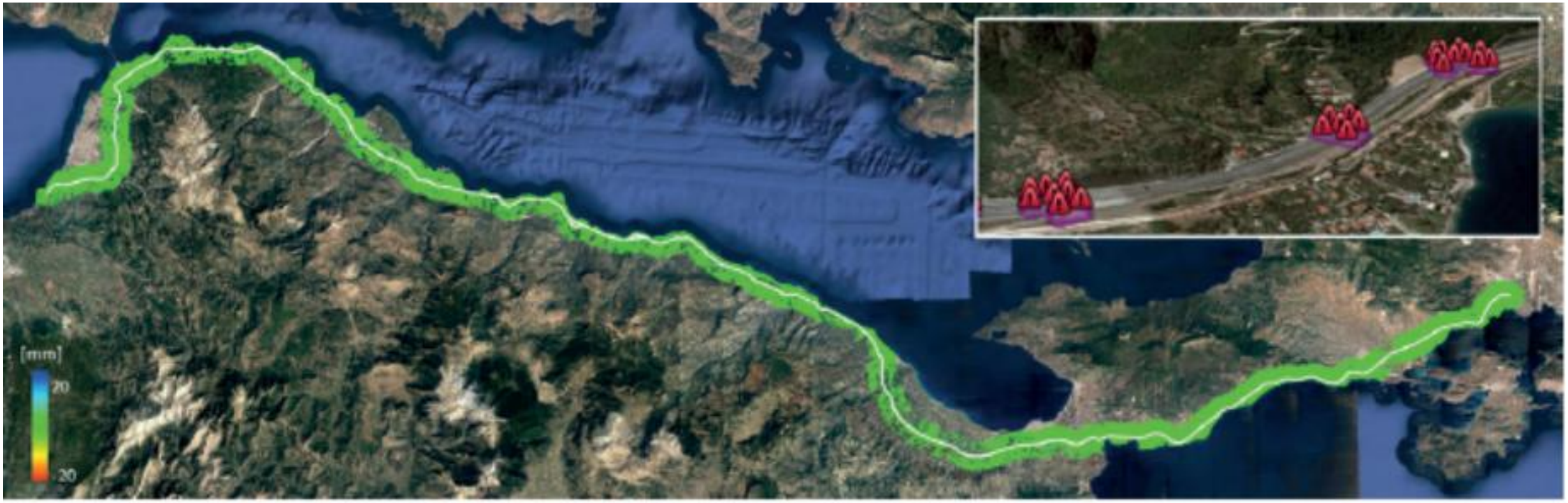
- InSAR Monitoring is the perfect tool to design long term maintenance programmes.
- It provides cost effective measurements and wide area coverage to assess structural health and detect displacement likely to cause damages to infrastructure.



Source: https://www.sixense-group.com/en/services/monitoring-testing/satellite-monitoring/atlas-insar-monitoring

Transport Corridors: Highways, Railways

- **Map ground** and **infrastructure motion** over hundreds of kilometers
- Quickly identify **active deformation areas** (unstable slopes, natural heave/subsidence, areas prone to landslide, etc.)
- Prioritize your on-site interventions to **protect transparent corridors, assets, and traffic**

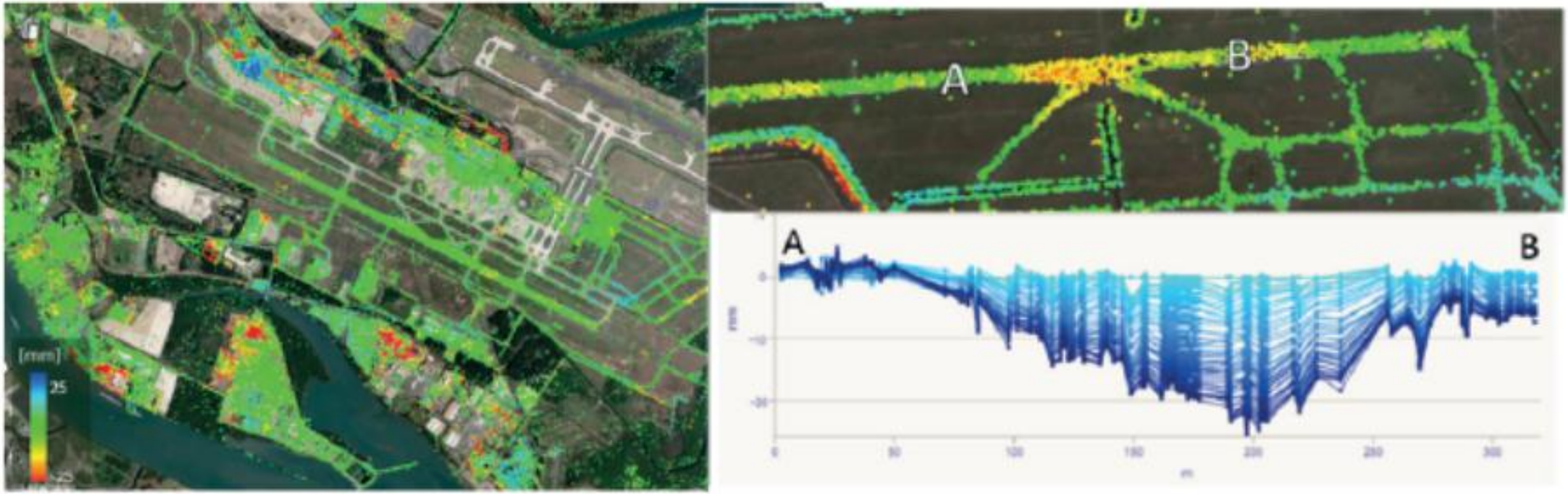


2 years of accumulated deformation on 200 km of motorways, and the filtering of areas affected by active deformation. Images Sentinel-1@ESA

Source: https://www.sixense-group.com/en/services/monitoring-testing/satellite-monitoring/atlas-insar-monitoring

Key Assets: Airports, Ports, Dams, Bridges

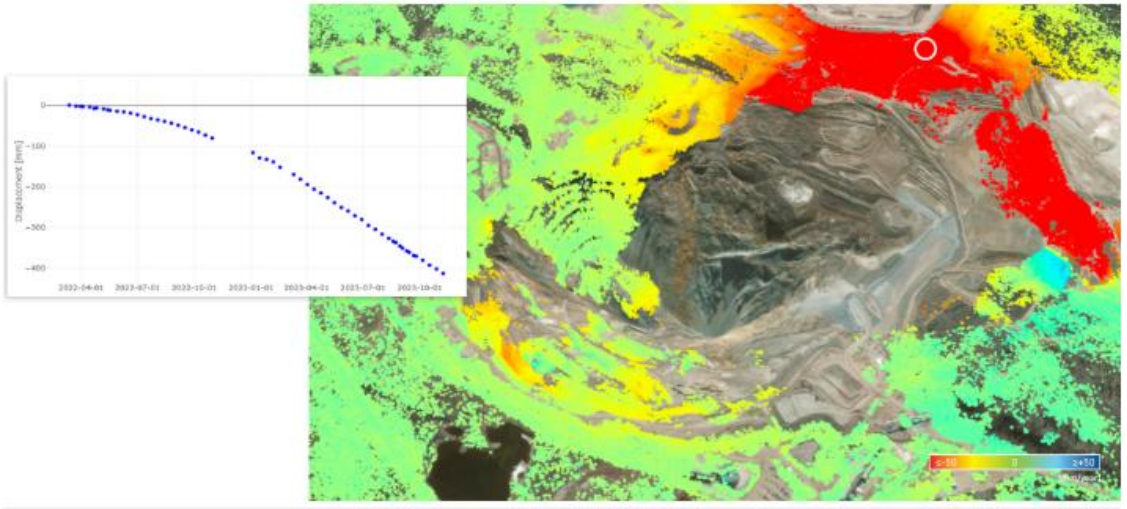
- Understand the **behavior** of your site and infrastructure over time
- Obtain **regular measurement data** to assist in the management of key assets and identify structural issues
- Optimize long-term **maintenance cost**

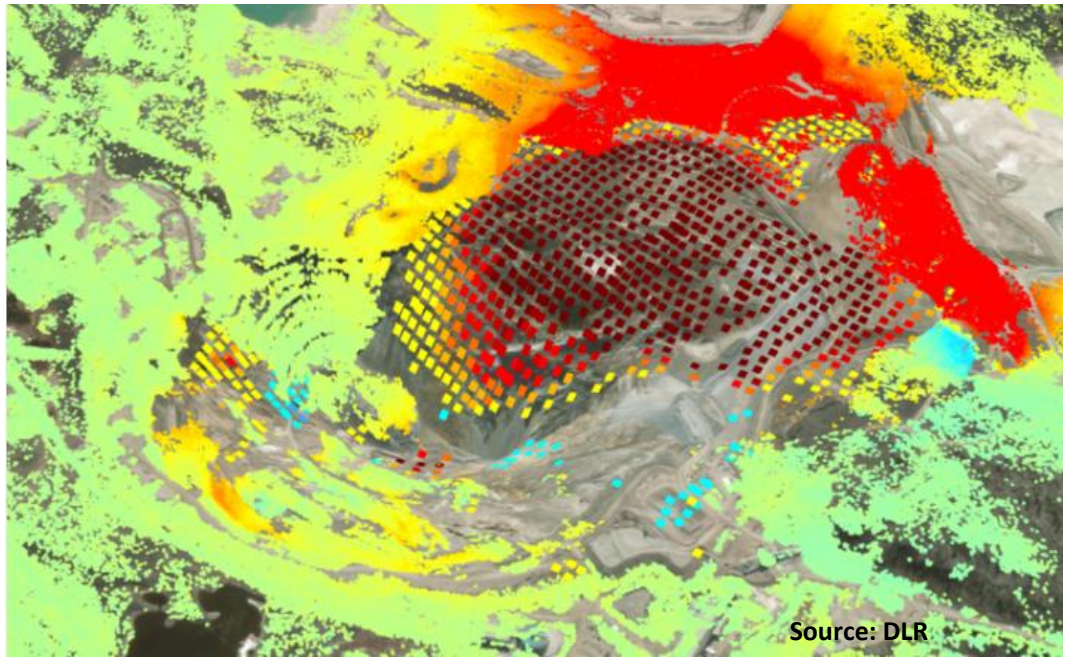


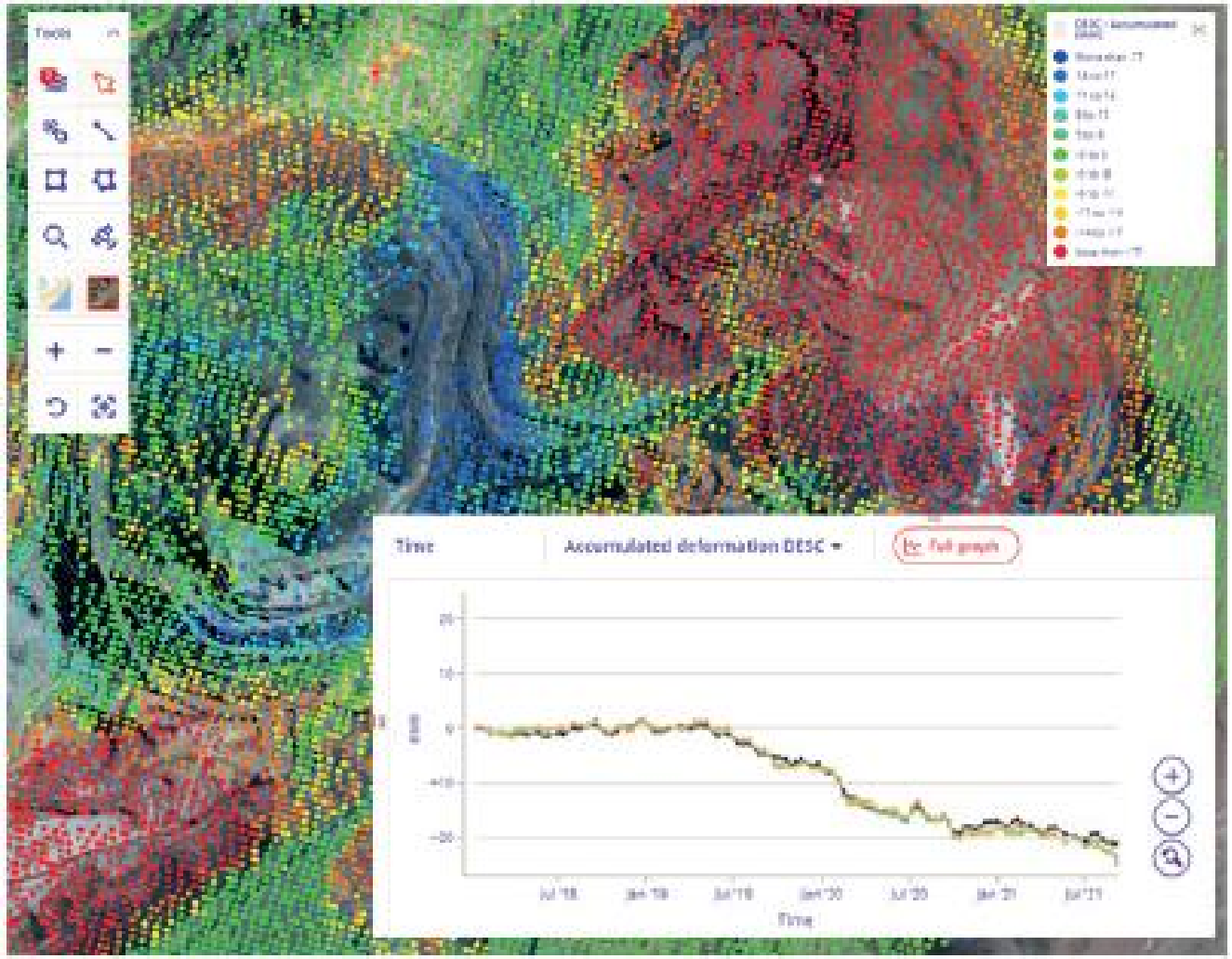
6 years of accumulated deformation at Brisbane Airport, the cross section presents spatial and temporal evolution of the motion along a section of the runway. Images TerraSAR-X@ Airbus

Source: https://www.sixense-group.com/en/services/monitoring-testing/satellite-monitoring/atlas-insar-monitoring

Mining Induced Land Subsidence



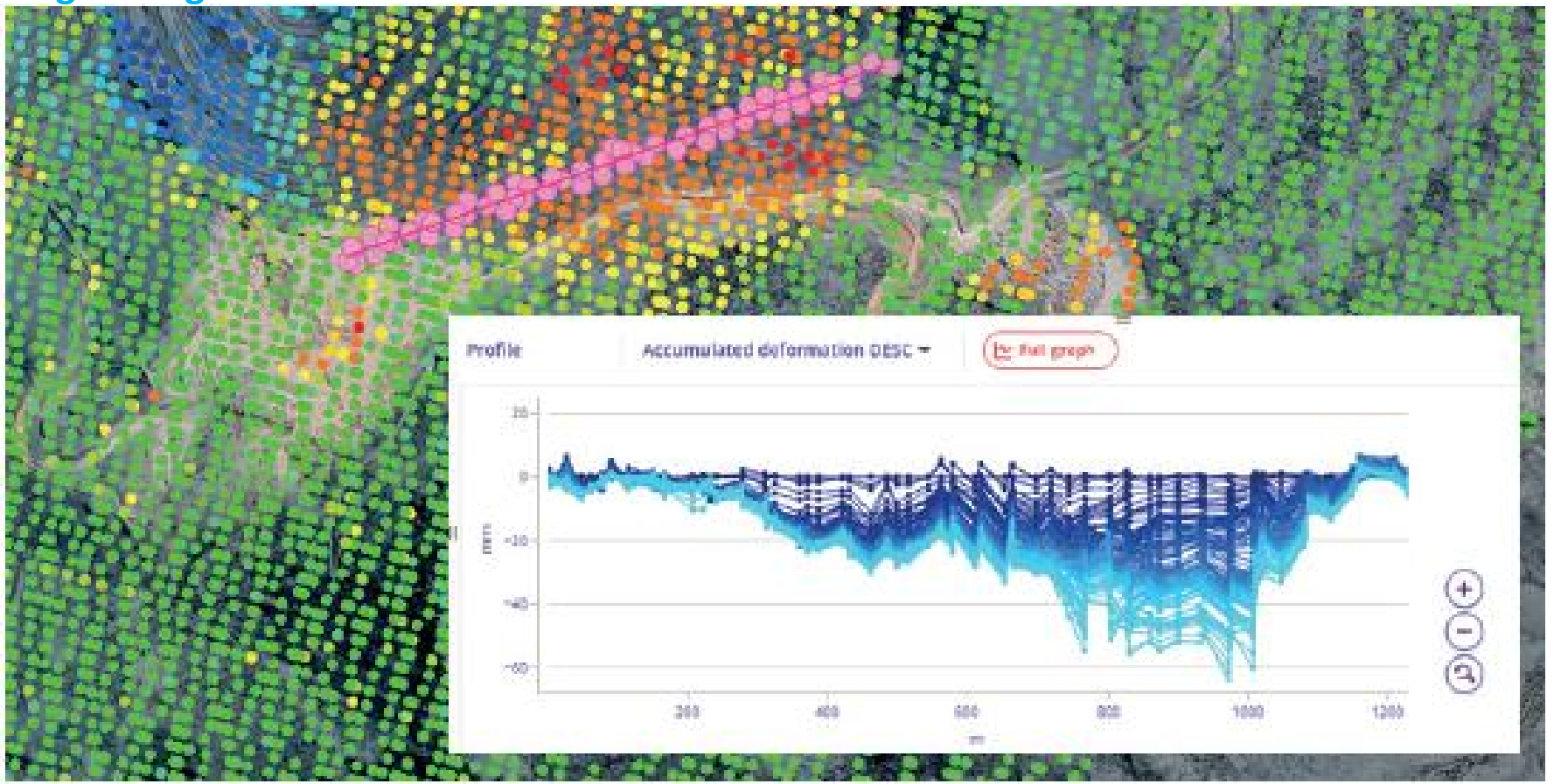




Source: https://www.sixense-group.com/en/services/monitoring-testing/satellite-monitoring/atlas-insar-monitoring

Open Pit and Underground Mining Operations

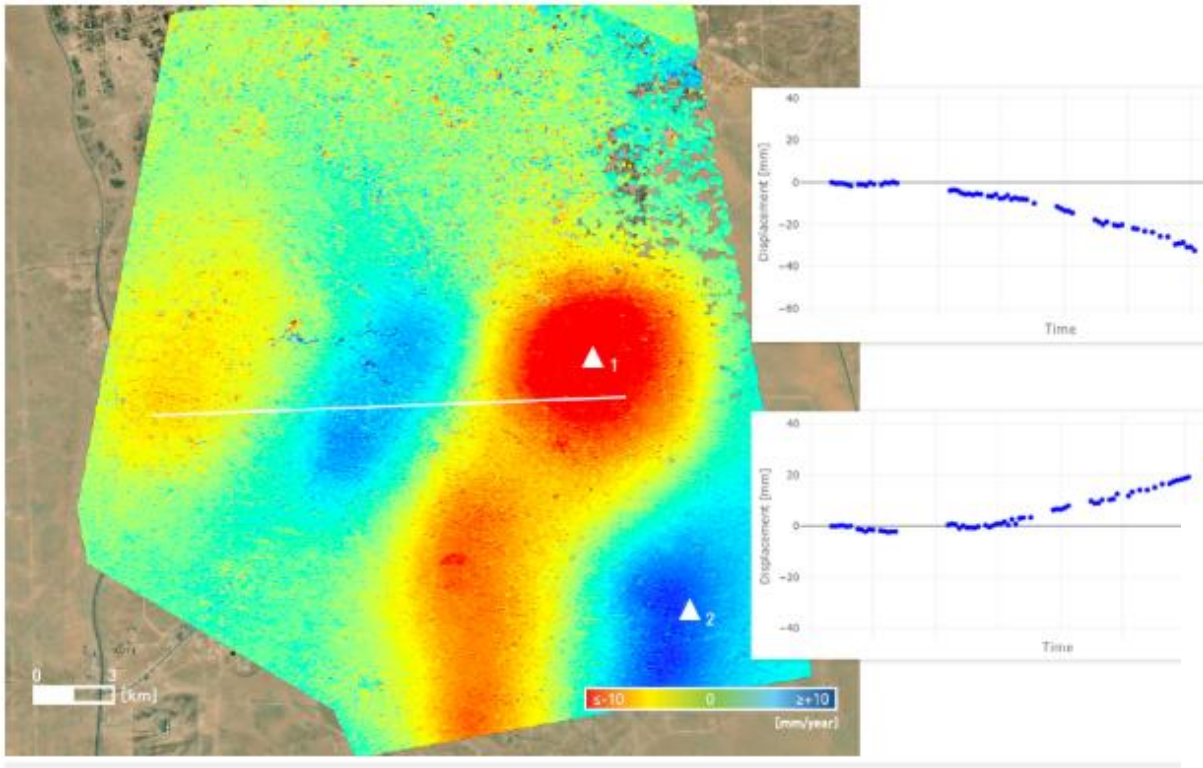
- **Monitoring** of **Slope stability** in open pit mines, quantification of movements and their progress over time
- Monitoring of settlements in heap leach pads, crushers, waste dumps, conveyor belts or other critical structures
- **Monitoring** of **subsidence** induced by underground mining activities such as block caving and long wall mining
- **Monitoring mining closure**



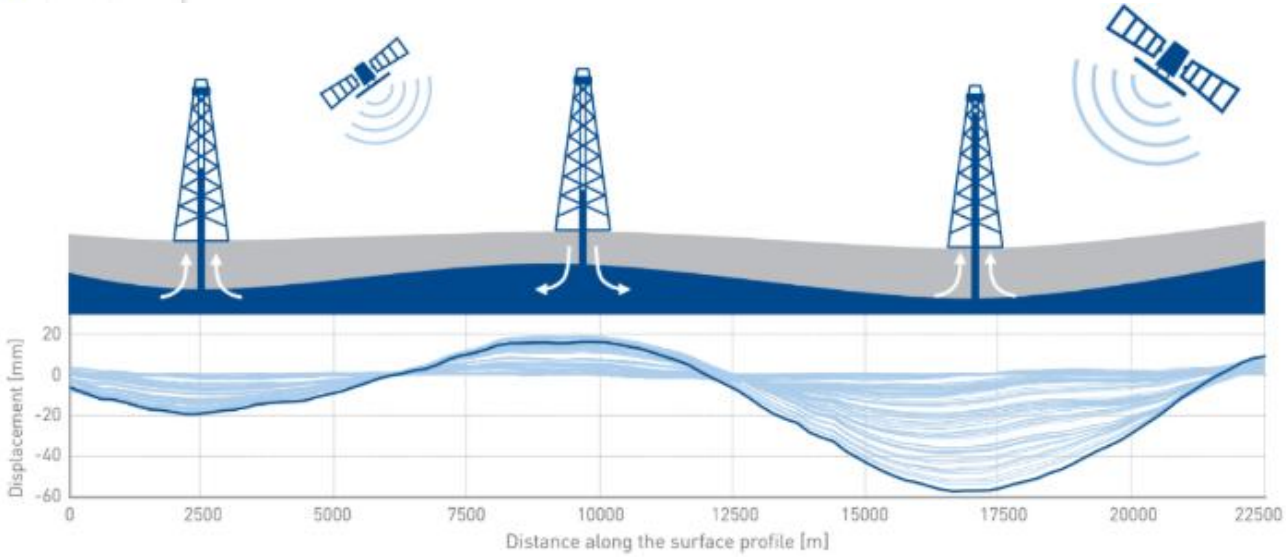
Source: https://www.sixense-group.com/en/services/monitoring-testing/satellite-monitoring/atlas-insar-monitoring

Enhanced Oil Recovery

- During Oil and gas production processes, fluid injection and extraction can induce pore pressure changes in the reservoir, potentially leading to ground deformation, which could affect wells and infrastructure



Source: https://site.tre-altamira.com/showcase/enhanced-oil-recovery/



Soil Moisture Retrieval

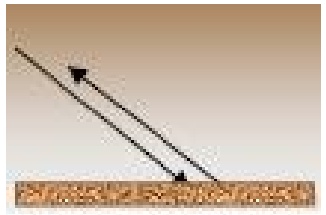
❖ Soil Moisture

- SM is the temporary storage of water within a shallow layer of the earth's upper surface.

$$Soil\ Moisture(Vol.\ \%) = \frac{Water\ Volume}{Total\ Volume\ of\ Soil\ Parcel}$$









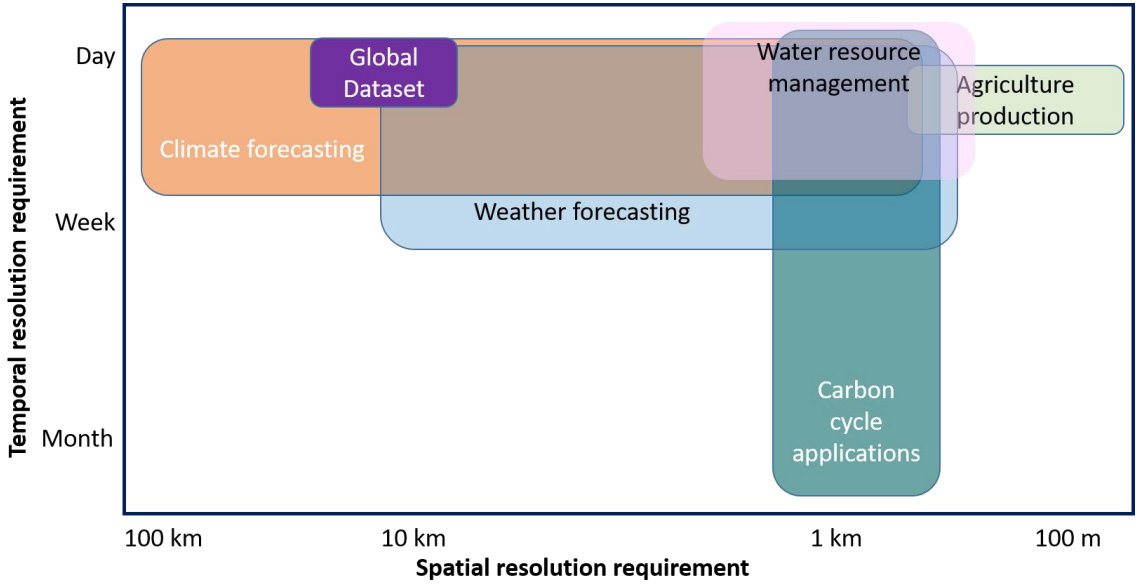


❖ Importance of Soil Moisture

- For each Kg of water on earth, only 1 mg is stored as soil moisture
- Soil Moisture exerts significant control over-
  - ✓ Hydrological processes
  - ✓ Ecological processes
  - ✓ Meteorological processes

❖ Factors influencing Soil Moisture

- ***Primary Factor:*** *rainfall, soil texture, vegetation, topography*
- ***Secondary Factors***
  - ✓ *System Parameters :* resolution, inc angle, polarisation, frequency
  - ✓ *Target Parameters :* Surface and land cover properties



Four Component Decomposition with Rotation of T_3

$$[T(\phi)] = \frac{P_s}{(1+|\beta|^2)}\begin{bmatrix}1 & \beta^* & 0\\ \beta & |\beta|^2 & 0\\ 0 & 0 & 0\end{bmatrix} + \frac{P_D}{(1+|\alpha|^2)}\begin{bmatrix}|\alpha|^2 & \alpha & 0\\ \alpha^* & 1 & 0\\ 0 & 0 & 0\end{bmatrix} + \frac{P_V}{4}\begin{bmatrix}2 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} + \frac{P_c}{2}\begin{bmatrix}0 & 0 & 0\\ 0 & 1 & \pm i\\ 0 & \mp i & 1\end{bmatrix}$$

- Rotating around line of sight through an orientation angle ϕ
- Rotation Matrix

$$[R(\phi)] = \begin{bmatrix}1 & 0 & 0\\ 0 & \cos 2\phi & \sin 2\phi\\ 0 & -\sin 2\phi & \cos 2\phi\end{bmatrix}$$

- Rotation Operation:

$$[T(\phi)] = R(\phi)TR(\phi)^{\dagger}$$

- **Surface scattering Power**

$$P_S = 0.5\, m_s{}^2 |R_H + R_V|^2 (1+|\beta|^2)$$

- **Surface roughness intensity Loss factor**

$$m_s = 2\cos(\theta_l{}^2).ks.kl.e^{\left\{\frac{1}{2}(kl\sin(\theta_l))^2\right\}}$$

- **Surface scattering mechanism ratio (Normalise Difference of Bragg Scattering in H & V Polarization)**

$$|\beta| = \left|\frac{R_H - R_V}{R_H + R_V}\right| \sin(2\psi_l) < 1$$

- **Bragg Scattering Coefficients**

$$R_H = \frac{\cos\theta - \sqrt{\varepsilon_r - \sin^2\theta}}{\cos\theta + \sqrt{\varepsilon_r - \sin^2\theta}}$$

$$R_V = \frac{(\varepsilon_r - 1)(\sin^2\theta - \varepsilon_r(1+\sin^2\theta))}{\left(\varepsilon_r\cos\theta + \sqrt{\varepsilon_r - \sin^2\theta}\right)^2}$$

Where, θ = Local Incidence Angle

- **Surface roughness parameter**

$$ks = \frac{1}{2\pi}\sin^{-1}\left(\frac{T_{XB22} - T_{XB33}}{T_{XB22} + T_{XB33}}\right)$$

- **Dihedral scattering intensity (Amplitude)**

$$P_D = 0.5 m_d{}^2 \left|R_{sh}R_{th} + R_{sv}R_{tv}.e^{i\phi}\right|^2 (1+|\alpha|^2)$$

- **Dihedral intensity loss factor**

$$\mathrm{m_d} = \mathrm{e}^{(-k^2 s^2 .\cos^2\theta_l)} \text{ for Gaussian ACF}$$
$$= \mathrm{e}^{(-\mathrm{ks}.\cos\theta_l)} \text{ for exponential ACF}$$

- **Dihedral scattering mechanism ratio**

$$|\alpha| = \left|\frac{R_{sh}R_{th} - R_{sv}R_{tv}.e^{i\phi}}{R_{sh}R_{th} + R_{sv}R_{tv}.e^{i\phi}}\right| < 1$$

- **Fresnel Scattering Coefficients (Soil/Trunk planes)**

$$R_{(s/t)h} = \frac{\cos\theta_{s/t} - \sqrt{\varepsilon_{s/t} - \sin^2\theta_{s/t}}}{\cos\theta_{s/t} + \sqrt{\varepsilon_{s/t} - \sin^2\theta_{s/t}}}$$

$$R_{(s/t)v} = \frac{\varepsilon_{s/t}.\cos\theta_{s/t} - \sqrt{\varepsilon_{s/t} - \sin^2\theta_{s/t}}}{\varepsilon_{s/t}.\cos\theta_{s/t} + \sqrt{\varepsilon_{s/t} - \sin^2\theta_{s/t}}}$$

Where, $\theta_t = \frac{\pi}{2} - \theta_s$ and ϕ = Differential phase

- **Surface roughness parameter**

$$ks = \frac{1}{2\pi}\sin^{-1}\left(\frac{T_{XF22} - T_{XF33}}{T_{XF22} + T_{XF33}}\right)$$

- Random (Uniform) Volume Dipole

$$\frac{P_V}{30}\begin{bmatrix}15 & 5 & 0\\ 5 & 7 & 0\\ 0 & 0 & 8\end{bmatrix}$$

- Vertical (Sine) Volume Dipole

$$\frac{P_V}{30}\begin{bmatrix}15 & -5 & 0\\ -5 & 7 & 0\\ 0 & 0 & 8\end{bmatrix}$$

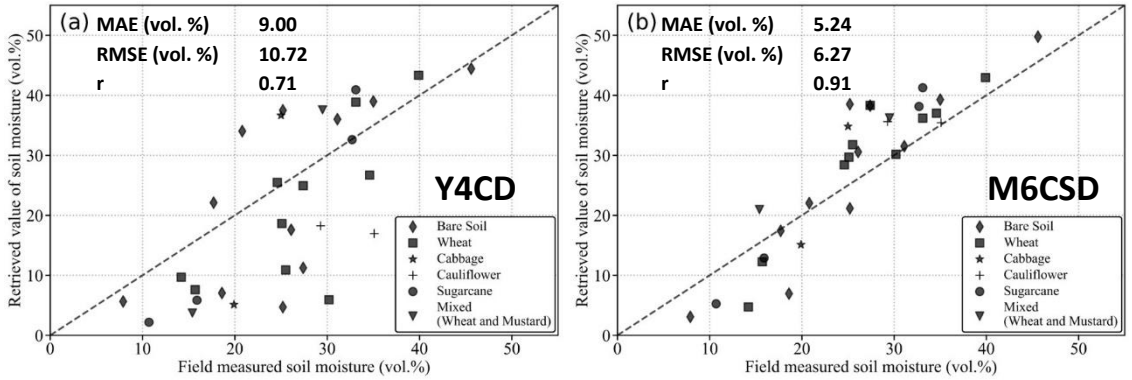
- Horizontal Volume (Cosine) dipole
- Extended Volume Component

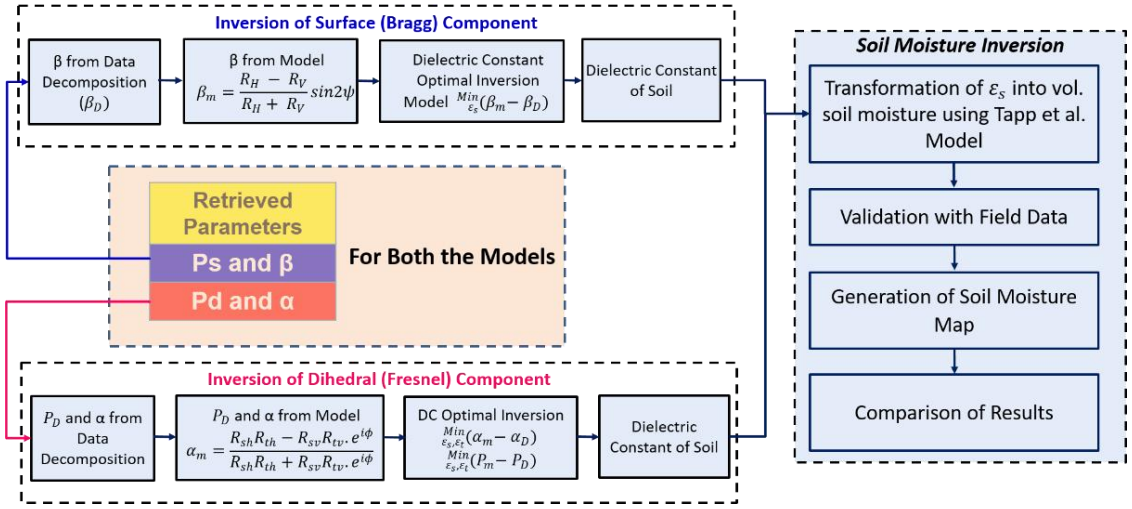
- Accounting for $Im(T_{23})$

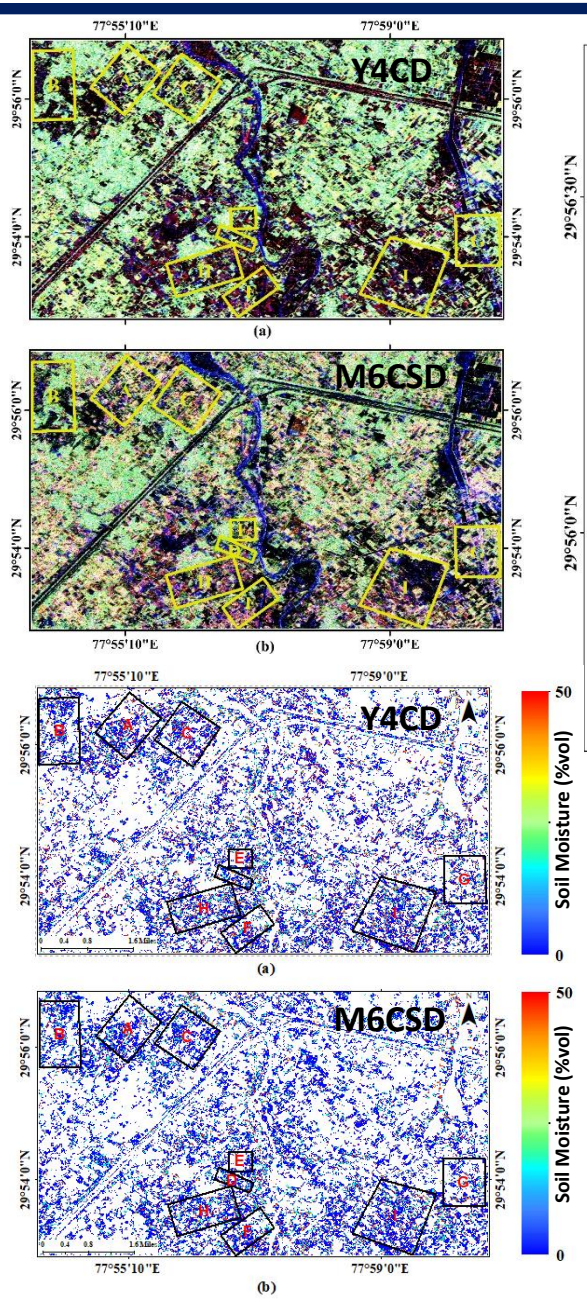
Soil Moisture Retrieval in Roorkee

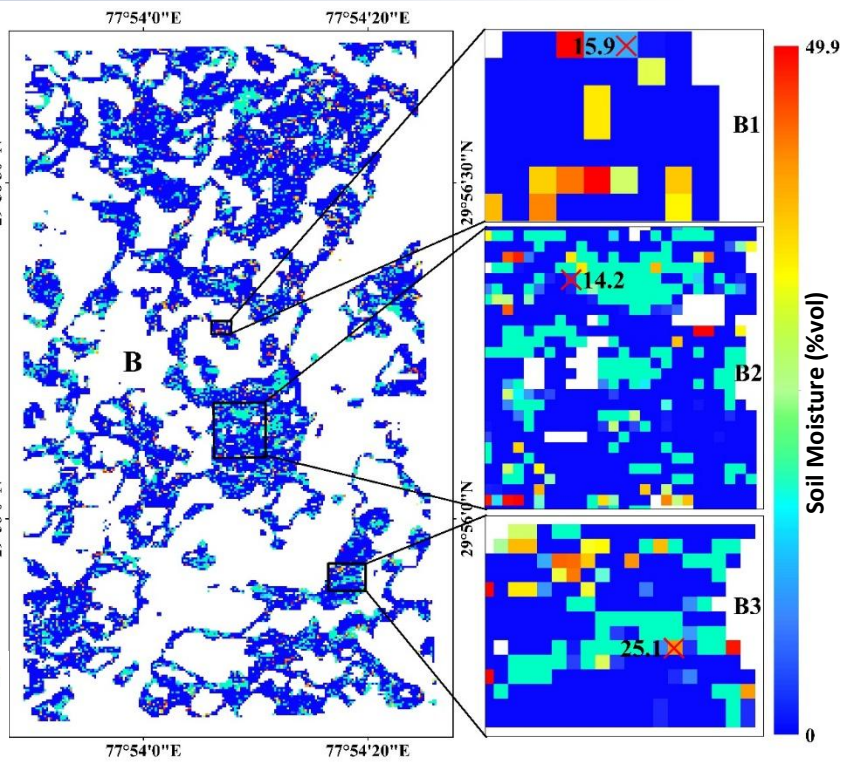
Objectives

- Identification of algorithm(s) for soil DC inversion with latest developments using polarimetric SAR data.
- Retrieval of surface soil moisture over agricultural land using full-pol SAR data and model based decomposition techniques.
- Comparison and validation of soil moisture retrieval results of polarimetric scattering decomposition models.









Highlights

- Soil moisture (vol.) was retrieved using Topp model.
- Performance indices indicate that M6CSD performed better than Y4CD.
- M6CSD model achieved 30.93% inversion rate while that of Y4CD was 24.34%.



Journal of the Indian Society of Remote Sensing (March 2022) 50(3):435–450
https://doi.org/10.1007/s12524-021-01471-x

RESEARCH ARTICLE

Model Based Four and Six Component Decompositions for Soil Moisture Retrieval

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Received: 7 January 2021 / Accepted: 29 November 2021 / Published online: 10 January 2022
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Thank You!